Highlights

Mesh-to-Mesh Transfer Errors in Multistep Unsteady Flow Simulations

Author names to be supplied

- Global mass correction does not restore a smeared wave profile.
- Shock-width diagnostics must exclude a smeared contact.
- Small transfer defects do not establish conservation during flow evolution.
- Local flux correction preserves mass but permits small relaxed-bound extrema.
- An error in an older BDF2 state changes the next-step lift.

Mesh-to-Mesh Transfer Errors in Multistep Unsteady Flow Simulations

Author names to be supplied[a]

[a]*Affiliation to be supplied, City, Country*

ARTICLE INFO

Keywords:
Solution transfer
Remeshing
Conservative interpolation
Galerkin projection
Backward differentiation
Finite-volume methods

ABSTRACT

When an unsteady flow simulation changes its mesh, transferring the solution can introduce errors that affect later flow evolution and aerodynamic loads. Multistep time integrators also require solutions from earlier times, so remeshing must transfer several stored states. Transfer is often assessed by field accuracy, the absence of new extrema, and preservation of domain integrals. These measures do not, however, directly describe how the restarted calculation will respond. This study asks how errors in the latest and older stored states influence that response, and what must be checked to assess transfer reliably.

We compare nearest-neighbor Taylor interpolation, barycentric interpolation, Galerkin projection, and control-volume transfer using static tests, repeated remeshing, and airfoil calculations. Repeated remeshing shows that restoring total mass does not recover a smeared wave profile. It also shows that preserving mass at each transfer does not ensure negligible subsequent mass drift: corrections that redistribute mass across the domain can alter open-boundary fluxes. Controlled restarts with the second-order backward differentiation formula (BDF2) show that an error in an older stored state changes next-step lift even when the latest state matches the reference. In the viscous airfoil test with the Spalart–Allmaras model, control-volume transfer best preserves the monitored integrals. Its root-mean-square drag departure from the fixed-mesh reference is the smallest on one mesh sequence and the largest on the other. The differences in drag between methods remain sensitive to the physical time step.

1. Introduction

Mesh adaptation allows an unsteady flow calculation to change its spatial resolution as the flow evolves. Every mesh change requires transferring the solution before time integration can continue. This transfer introduces another source of numerical error, in addition to spatial discretization and time integration, and repeated transfers can accumulate errors [1–3]. For an aerodynamic calculation, the practical concern is how these changes affect the subsequent flow and the predicted forces.

Transfer quality is commonly described through accuracy, boundedness, and conservation. Accuracy measures how closely the transferred field represents the source field; approximation order describes how that error decreases under mesh refinement for smooth data. Boundedness asks whether transfer creates values outside the source range. Conservation asks whether selected integrals or moments are preserved. Common-refinement and Galerkin methods provide accurate conservative transfer between nonmatching meshes [4, 5]. Constrained formulations can also preserve norms and weak incompressibility [6]. Conservative remapping integrates a reconstructed source field over target cells [7–10], while conservative Galerkin transfer has been used in high-order implicit adaptation with bounds near discontinuities [3, 11].

These properties describe the transfer itself, but do not directly determine the quality of the calculation after it restarts. Preserving total mass does not determine where that mass is located, and suppressing new extrema can modify both accuracy and conservation. The subsequent flow also depends on the spatial discretization, boundary conditions, and time integrator. Related shallow-water studies show that preserving a discrete equilibrium can suppress disturbances introduced by transfer [12, 13]. This connection between transfer and subsequent evolution motivates examining the complete restart, rather than judging only the field immediately after transfer.

Multistep time integration makes this distinction particularly important. It uses the latest solution together with solutions stored from earlier times, which we call history states. After remeshing, these states have been transferred from another mesh rather than computed by earlier steps on the target mesh. An error in an older state can therefore change the next time-step equation even when the latest state agrees with a target-mesh reference. Limiting, corrections

ORCID(s):

for positive density and internal energy, and the reapplication of boundary conditions can further change the states supplied to that equation.

The central question of this paper is whether conventional transfer-quality measures are sufficient to assess the subsequent unsteady flow and aerodynamic loads. We ask how errors in the latest and older history states contribute to that response, and which additional checks are needed to interpret a comparison between transfer methods. To address these questions, we examine both the changes introduced during transfer and the flow computed after the restart. Controlled restarts separate the effects of errors in different history states; repeated remeshing shows how transfer and flow evolution interact over many events. Time-step and iterative-convergence checks assess whether observed load differences can support a ranking of the transfer methods.

We compare nearest-neighbor Taylor interpolation (NN), barycentric interpolation (BAR), continuous Galerkin projection (GP), and conservative control-volume transfer (CV) in a vertex-centered finite-volume solver. The unsteady calculations use the second-order backward differentiation formula (BDF2). An exact identity relates errors in the two stored BDF2 states to the change in the time-step residual. We distinguish this identity from the additional assumptions needed to predict the computed state and load response. Global integral correction and local flux correction are examined separately to establish what restoring an integral can, and cannot, resolve.

The numerical tests progress from smooth and discontinuous static fields to repeated remeshing of a Riemann problem, controlled airfoil restarts, and repeated-remeshing airfoil calculations. Unsteady Reynolds-averaged Navier–Stokes (RANS) calculations on a stationary airfoil with the Spalart–Allmaras (SA) model extend the assessment to turbulence transport and viscous loads. All tests use prescribed remeshing on fixed physical domains and independently computed reference solutions; they do not assess the choice of an adaptive mesh. Section 2 describes the transfer methods. The BDF2 analysis then connects history-state errors to the restart response, and each test-case subsection presents its setup, results, and interpretation together.

2. Transferring a solution between meshes

The transfer methods differ in how they compute values on the target mesh. NN and BAR interpolate from nearby source values; GP and CV solve coupled systems based on integrals. Each method can then apply limiting, boundary conditions, and corrections for positive density and internal energy. We distinguish the field before these operations from the field passed to the flow solver. Method suffixes give nominal accuracy order rather than polynomial degree: BAR2 is second order, and GP2 and CV2 use piecewise-linear fields.

2.1. Stored variables and reconstructed derivatives

The source mesh A and target mesh B cover the same discretized domain Ω in spatial dimension d. A mesh M has n_M solver vertices. Vertex i has position $\boldsymbol{x}_i^M$ and a surrounding median-dual control volume C_i^M of positive measure $w_i^M = |C_i^M|$. The symbol $|C|$ denotes area or volume, and $\mathrm{d}x$ the corresponding integration measure. For mesh M, vertex indices range from 1 to n_M.

The solver state U_M is stacked vertex by vertex, with n_U transported variables per vertex. A superscript T means transpose. For a scalar component ϕ, $\phi_M = (\phi_{M,1}, \dots, \phi_{M,n_M})^T$ collects its vertex values, and its discrete integral is

$$Q_M(\phi_M) = \sum_{i=1}^{n_M} w_i^M \phi_{M,i}. \tag{1}$$

For the full state U_M, $Q_M(U_M)$ means the vector of these integrals, one per transported component.

For a nonzero source integral, define the signed relative integral change as $e_Q(\phi_A, \phi_B) = [Q_B(\phi_B) - Q_A(\phi_A)]/Q_A(\phi_A)$. Static and RANS transfer diagnostics use this normalization.

Mean-flow transfer acts on density ρ, momentum density $\rho\boldsymbol{u}$, and total energy per unit volume ρE. Here $\boldsymbol{u}$ is velocity, with Cartesian components (u, v) in two dimensions, and E is total energy per unit mass. All methods supply values at the solver vertices, not control-volume averages.

Higher-order methods also need local slopes and curvature. Weighted least squares reconstructs them from neighboring source values:

$$\phi_j - \phi_i \simeq \boldsymbol{g}_i \cdot \boldsymbol{r}_{ij} + \tfrac{1}{2}\boldsymbol{r}_{ij}^T \mathcal{H}_i \boldsymbol{r}_{ij}, \qquad \boldsymbol{r}_{ij} = \boldsymbol{x}_j - \boldsymbol{x}_i. \tag{2}$$

The vector $\boldsymbol{r}_{ij}$ joins source vertices i and j; $\boldsymbol{g}_i$ and $\mathcal{H}_i$ approximate the gradient and Hessian of ϕ at i. A linear fit omits the Hessian. For smooth data on a well-conditioned, full-rank scaled stencil, with controlled geometry and characteristic local mesh spacing h, the linear fit has an $O(h)$ gradient error. A quadratic fit has $O(h^2)$ gradient and $O(h)$ Hessian errors. Anisotropy, weighting, and poor conditioning can reduce this accuracy [14–17]. Third-order methods use the quadratic reconstruction, and their achieved order is checked numerically.

2.2. Nearest-neighbor and barycentric interpolation

The superscript on $\mathcal{I}^m_{AB}$ identifies a raw interpolation formula for method m, before limiting and state corrections. Let $i_{\mathrm{NN}}(j)$ be the nearest source vertex to target point $\boldsymbol{x}^B_j$, and let $\boldsymbol{r}_j = \boldsymbol{x}^B_j - \boldsymbol{x}^A_{i_{\mathrm{NN}}(j)}$ be their displacement. Nearest-neighbor Taylor transfer gives

$$(\mathcal{I}^{\mathrm{NN1}}_{AB}\phi_A)_j = \phi_{A,i_{\mathrm{NN}}(j)}, \tag{3}$$

$$(\mathcal{I}^{\mathrm{NN2}}_{AB}\phi_A)_j = \phi_{A,i_{\mathrm{NN}}(j)} + \boldsymbol{g}_{i_{\mathrm{NN}}(j)} \cdot \boldsymbol{r}_j, \tag{4}$$

$$(\mathcal{I}^{\mathrm{NN3}}_{AB}\phi_A)_j = \phi_{A,i_{\mathrm{NN}}(j)} + \boldsymbol{g}_{i_{\mathrm{NN}}(j)} \cdot \boldsymbol{r}_j + \tfrac{1}{2}\boldsymbol{r}_j^T \mathcal{H}_{i_{\mathrm{NN}}(j)} \boldsymbol{r}_j. \tag{5}$$

NN1 copies the nearest source value, NN2 adds its local slope, and NN3 also includes curvature. For $\|\boldsymbol{r}_j\|_2 = O(h)$, where $\|\cdot\|_2$ is the Euclidean norm, and sufficiently accurate derivatives, the smooth-field errors are $O(h)$, $O(h^2)$, and $O(h^3)$. Changing the nearest source vertex can make the interpolated value discontinuous as a function of target position.

Barycentric interpolation uses the source triangle or tetrahedron containing the target point. A conforming simplicial subdivision supplies this search geometry on other element types without adding control-volume vertices. In dimension d, the containing simplex has vertices $\boldsymbol{x}^A_{i_r}$, $r = 1, \ldots, d+1$. Its barycentric coordinates are nonnegative weights that express the target position:

$$\boldsymbol{x}^B_j = \sum_{r=1}^{d+1} \lambda_r \boldsymbol{x}^A_{i_r}, \qquad \sum_r \lambda_r = 1, \qquad \lambda_r \geq 0. \tag{6}$$

The same weights define BAR2 and its gradient-corrected extension:

$$(\mathcal{I}^{\mathrm{BAR2}}_{AB}\phi_A)_j = \sum_r \lambda_r \phi_{A,i_r}, \tag{7}$$

$$(\mathcal{I}^{\mathrm{BAR3}}_{AB}\phi_A)_j = \sum_r \lambda_r \left[\phi_{A,i_r} + \tfrac{1}{2}\boldsymbol{g}_{i_r} \cdot (\boldsymbol{x}^B_j - \boldsymbol{x}^A_{i_r})\right]. \tag{8}$$

BAR2 reproduces affine fields, meaning a linear function plus a constant, and introduces no new maximum or minimum relative to the containing-simplex vertex values, because its nonnegative weights sum to one. For a quadratic field with exact vertex gradients, the half-gradient term cancels the interpolation remainder, giving BAR3 third-order consistency. Shared values and gradients make the interpolant continuous across conforming simplex interfaces. Neither method constrains the source-to-target integral.

For completeness, let q be a quadratic field with constant Hessian $\mathcal{H}_q$, let $\boldsymbol{x} = \boldsymbol{x}^B_j$, and set $\boldsymbol{d}_r = \boldsymbol{x} - \boldsymbol{x}^A_{i_r}$. Exact Taylor expansion gives

$$q(\boldsymbol{x}^A_{i_r}) = q(\boldsymbol{x}) - \nabla q(\boldsymbol{x}) \cdot \boldsymbol{d}_r + \tfrac{1}{2}\boldsymbol{d}_r^T \mathcal{H}_q \boldsymbol{d}_r,$$

$$\nabla q(\boldsymbol{x}^A_{i_r}) = \nabla q(\boldsymbol{x}) - \mathcal{H}_q \boldsymbol{d}_r.$$

Substituting these expressions into Eq. (8) gives $q(\boldsymbol{x}) - \frac{1}{2}\nabla q(\boldsymbol{x}) \cdot \sum_r \lambda_r \boldsymbol{d}_r = q(\boldsymbol{x})$, because the barycentric coordinate identity makes the sum zero. For a smooth field on a shape-regular simplex, the Taylor remainder is $O(h^3)$; an $O(h^2)$ gradient error also contributes $O(h^3)$ when multiplied by the $O(h)$ displacement.

2.3. Galerkin projection and cross-mesh integration

Galerkin projection chooses the target field that best matches the reconstructed source field in the L_2 norm [4, 5]. This transfer uses finite-element basis functions, while the flow equations here remain finite volume. It solves for the target values together rather than interpolating each vertex independently. Let ϕ^h_A be the piecewise-polynomial

source field and V_B^ℓ the space of continuous Lagrange functions of polynomial degree ℓ on the target simplicial mesh. The superscript h denotes the piecewise-polynomial field built from the source vertex values; GP3 also uses their reconstructed derivatives. For scalar fields f_1, f_2, the inner product is $(f_1, f_2)_{L_2(\Omega)} = \int_\Omega f_1 f_2 \, \mathrm{d}x$; the associated norm is $\|f_1\|_{L_2(\Omega)} = (f_1, f_1)^{1/2}_{L_2(\Omega)}$. The target field $\phi_B^h \in V_B^\ell$ satisfies

$$(\phi_B^h - \phi_A^h, \xi_B)_{L_2(\Omega)} = 0 \quad \text{for every } \xi_B \in V_B^\ell. \tag{9}$$

Thus the transfer error has zero integral when weighted by any target test function. This condition gives the best L_2 approximation of ϕ_A^h in V_B^ℓ [4, 5]. The test function ξ_B is arbitrary in that target space. Let N_j^B be its nodal basis functions and N_B^{FE} their number. The target coefficient vector $z_B = (z_{B,1}, \ldots, z_{B,N_B^{\mathrm{FE}}})^T$ defines $\phi_B^h = \sum_{j=1}^{N_B^{\mathrm{FE}}} z_{B,j} N_j^B$. Taking $\xi_B = N_i^B$ in Eq. (9) and substituting the basis expansion gives $\sum_j (\int_\Omega N_i^B N_j^B \, \mathrm{d}x) z_{B,j} = \int_\Omega N_i^B \phi_A^h \, \mathrm{d}x$. In matrix form,

$$H_B z_B = b, \qquad (H_B)_{ij} = \int_\Omega N_i^B N_j^B \, \mathrm{d}x, \qquad b_i = \int_\Omega N_i^B \phi_A^h \, \mathrm{d}x. \tag{10}$$

The consistent Galerkin mass matrix H_B is sparse and positive definite, and b is the cross-mesh load vector. The matrix couples basis functions that share target elements.

The two meshes generally do not share element boundaries. A target element can overlap several source elements, across which the source polynomial changes. Integration must account for these intersections:

$$b_i = \sum_{K_B} \sum_{K_A \colon |K_A \cap K_B| > 0} \int_{K_A \cap K_B} N_i^B \phi_A^h \, \mathrm{d}x. \tag{11}$$

Here K_A and K_B are source and target simplices, and $|K_A \cap K_B|$ is their intersection measure. Their intersections form a common refinement (Figure 1a). Both fields have one polynomial expression on each intersection, allowing exact polynomial quadrature: degree two for P^1/P^1 transfer and degree four for P^2/P^2 transfer on affine simplices [18, 19]. The intersections must cover the domain without gaps or overlap. This geometric integration and the coupled solve distinguish GP from point interpolation.

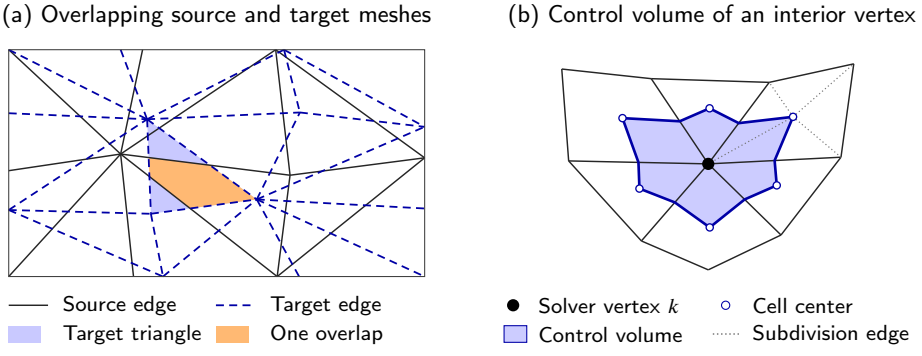


Figure 1: Geometry used in cross-mesh integration. (a) One target triangle crosses several source triangles. The darker polygon is its intersection with one source triangle and gives one region for integrating the Galerkin right-hand side. (b) The six triangles and quadrilaterals around vertex k each contribute a shaded piece to its median-dual control volume C_k^M. The blue boundary joins edge midpoints to cell centers. Dotted lines in one quadrilateral show the triangles used for reconstruction. Only the original mesh vertices carry solver unknowns. Both panels are illustrative mesh patches.

GP2 uses piecewise-linear (P^1) source and target fields. GP3 reconstructs a quadratic (P^2) source field, including internal edge values, and solves for a complete quadratic target field. The solver receives only the original-vertex values. Restricting the quadratic field in this way can change its finite-volume integral, as discussed in Sec. 2.5. Here P^ℓ means polynomials of total degree at most ℓ, while Q^ℓ permits degree at most ℓ in each coordinate. Mixed-element meshes use P^ℓ on simplicial subdivisions. The coefficient vector z_B contains every finite-element degree of freedom; the solver vector ϕ_B retains only its original-vertex entries. Under smoothness and mesh-stability assumptions, the nominal orders are two and three; the accuracy of the vertex values is tested separately.

2.4. Conservative control-volume transfer

The CV method matches source and target field integrals over each target control volume. The solver computes a total by multiplying each vertex value by its control-volume area (or volume in three dimensions) and summing over the mesh. CV needs a field that can also be integrated over pieces of cells. We construct that field so that these two ways of computing the total agree. For an element K of mesh M, the portion of vertex i's control volume inside it is $C_i^M \cap K$. Both meshes represent the field as $\phi_M^h = \sum_k \phi_{M,k}\psi_k^M$. On triangles and tetrahedra, $\psi_k^M = N_k^M$ is the usual P^1 hat function. A quadrilateral is divided into triangles by joining its vertices to their average center (Figure 1b). The value at that center is a weighted combination of the original nodal values, rather than an independent unknown. Nonsimplicial three-dimensional elements use analogous centered subdivisions. These dependent values are chosen so that

$$\int_K \psi_k^M \,\mathrm{d}x = |C_k^M \cap K| \qquad \text{for every element } K \text{ and vertex } k \text{ of } M. \tag{12}$$

For a quadrilateral, vertex k contributes to the value at the center with weight $|C_k^M \cap K|/|K|$. To see why Eq. (12) matters, multiply it by the stored value $\phi_{M,k}$ and sum over the vertices of K. The resulting field integral is $\int_K \phi_M^h \,\mathrm{d}x = \sum_{k\in K} |C_k^M \cap K|\phi_{M,k}$. Summing over all elements gives exactly the solver total $Q_M(\phi_M)$. The reconstructed field and the solver thus have the same domain integral whenever Eq. (12) holds.

The source field interpolates nodal solver values continuously; it is not reconstructed from source-cell averages. Its domain integral equals $Q_A(\phi_A)$, but its integral over an individual source control volume need not equal $w_i^A\phi_{A,i}$. Before limiting and boundary treatment, CV preserves the source integral over the whole domain and matches the reconstructed field integral over each target control volume. It does not preserve each source nodal content.

On parallelograms, the center weights give the arithmetic vertex average and reproduce affine fields. On quadrilaterals that remain distorted, they do not: the weights enforce Eq. (12) at the expense of exact affine reproduction.

The n_B target vertex values are defined by

$$\int_{C_j^B} \left(\phi_B^h - \phi_A^h\right) \,\mathrm{d}x = 0, \qquad j = 1, \ldots, n_B. \tag{13}$$

Equation (13) determines the target nodal values by matching one integral per target control volume. In finite-element terms, it is a Petrov–Galerkin condition: the trial functions are ψ_k^B, while the test functions equal one inside a target control volume and zero outside. Intersection-based P^1-conservative interpolation [9, 10] instead preserves integrals over target elements, limits an element-wise linear reconstruction, and obtains vertex values by explicit volume-weighted averaging over incident elements. Matching the solver control volumes directly requires a sparse solve:

$$\widetilde{H}_B\phi_B = \tilde{b}, \qquad (\widetilde{H}_B)_{jk} = \int_{C_j^B} \psi_k^B \,\mathrm{d}x, \qquad \tilde{b}_j = \int_{C_j^B} \phi_A^h \,\mathrm{d}x. \tag{14}$$

The functions ψ_k^B sum to one, giving row sums w_j^B; Eq. (12) gives column sums w_k^B. The entries are nonnegative and couple the same elements as H_B. They depend only on the target geometry and are evaluated in closed form. The matrix is symmetric on triangles, strictly convex quadrilaterals, and tetrahedra. Vertex permutations exchange the median-dual pieces of a simplex; on a quadrilateral, each center weight is proportional to the corresponding median-dual piece.

The right-hand-side vector $\tilde{b}$ contains integrals of the piecewise-linear source field over target control volumes. It is assembled from intersections of source subelements and target median-dual pieces:

$$\tilde{b}_j = \sum_{K_B \ni j} \sum_{K_A\colon |K_A\cap K_B|>0} \int_{K_A\cap K_B\cap C_j^B} \phi_A^h \,\mathrm{d}x, \tag{15}$$

Here K_B ranges over target elements incident to j, and K_A over source subelements. In these calculations, the meshes cover the same polygonal domain and the uncovered measure is below 10^{-20} of the domain. Equation (14) is solved to a relative residual tolerance of 10^{-15}. Its unknowns are nodal values, not the control-volume averages $\tilde{b}_j/w_j^B$.

Summing Eq. (13) over the target control volumes, which tile Ω, and using Eq. (12) gives

$$
\begin{aligned}
Q_B(\phi_B) &= \sum_{j=1}^{n_B} \int_{C_j^B} \phi_B^h \, \mathrm{d}x = \sum_{j=1}^{n_B} \tilde{b}_j \\
&= \int_\Omega \phi_A^h \, \mathrm{d}x = Q_A(\phi_A).
\end{aligned}
\tag{16}
$$

to within integration and solve tolerances. Locally, Eq. (13) equates $\int_{C_j^B} \phi_B^h \, \mathrm{d}x$ and $\int_{C_j^B} \phi_A^h \, \mathrm{d}x$, integrating the target and source reconstructed fields over the same target region. This integral differs from the nodal content $w_j^B \phi_{B,j}$ by $(\widetilde{H}_B \phi_B)_j - w_j^B \phi_{B,j}$. These differences sum to zero but need not vanish individually. Transferring back to the same mesh, $B = A$, gives $\tilde{b} = \widetilde{H}_A \phi_A$ and recovers ϕ_A. If both representations reproduce affine fields, uniqueness of Eq. (14) also makes their transfer exact. CV2 is therefore nominally second order on simplicial meshes and quadrilaterals that remain parallelograms under refinement. Persistent quadrilateral distortion reduces the attainable order. No higher-order CV method is used.

On simplicial meshes, CV2 and GP2 use the same continuous P^1 source and target fields. Their test functions differ. GP2 weights the error by continuous hat functions and gives the best L_2 approximation. Choosing the constant function or a coordinate as the test function in Eq. (9) explains why the unconstrained projection preserves the field integral and its coordinate moments. CV2 matches integrals over target control volumes and preserves the global finite-volume integral before limiting and boundary treatment. It generally preserves neither coordinate moments nor the minimum L_2 error. Both require a coupled solve: changing the mesh in one region can alter transferred values elsewhere. On quadrilaterals, their field representations also differ. GP2 triangulates without added vertices; CV2 uses a centered subdivision with dependent center values.

The symmetry established above and strict diagonal dominance make the CV recovery matrix positive definite on triangular, strictly convex quadrilateral, and tetrahedral meshes. These properties do not generally extend to mixed three-dimensional cells. All numerical comparisons here are two-dimensional; persistent quadrilateral distortion is the main limitation on CV accuracy.

2.5. Field integrals and the solver integral

For GP2, the field integral and the finite-volume integral agree when $w_i^M = \int_\Omega N_i^M \, \mathrm{d}x$ on both meshes. This holds on simplicial meshes. It generally fails when a quadrilateral mesh is triangulated, because the hat-function integrals differ from the median-dual measures. CV2 enforces the corresponding equality through Eq. (12) for any subdivision satisfying that identity. GP3 adds another difference: reconstructing a quadratic source field and then retaining only target vertices changes the representation. The next example shows that even an exact quadratic projection can then have a second-order finite-volume integral error. Third-order point-value accuracy does not imply third-order accuracy of the solver's integral.

The difference can arise even when every transferred vertex value is exact (Figure 2). On a uniform partition of $[0, 1]$ into N_{seg} segments of length $h = 1/N_{\mathrm{seg}}$, let $f(x) = x^2$ and f_M be its vertex values. The median-dual weights are $h/2$ at the endpoints and h at interior vertices. Equation (1) then gives

$$
\begin{aligned}
Q_M(f_M) &= h \left[\tfrac{1}{2} f(0) + \sum_{i=1}^{N_{\mathrm{seg}}-1} (ih)^2 + \tfrac{1}{2} f(1) \right] \\
&= \frac{h^3}{6} (N_{\mathrm{seg}} - 1) N_{\mathrm{seg}} (2N_{\mathrm{seg}} - 1) + \frac{h}{2} = \frac{1}{3} + \frac{h^2}{6}.
\end{aligned}
\tag{17}
$$

Here $\int_0^1 f(x) \, \mathrm{d}x = 1/3$, and $\sum_{i=1}^{N-1} i^2 = (N-1)N(2N-1)/6$ with $N = N_{\mathrm{seg}}$ gives the displayed sum. Thus exact vertex values still give a finite-volume integral error of $h^2/6$. For transfer of the same quadratic between two uniform meshes of spacings h_A and h_B, the integral change is $Q_B(f_B) - Q_A(f_A) = (h_B^2 - h_A^2)/6$, where f_A, f_B are the sampled fields. This separates a source-to-target integral change from either mesh's error relative to the continuum integral.

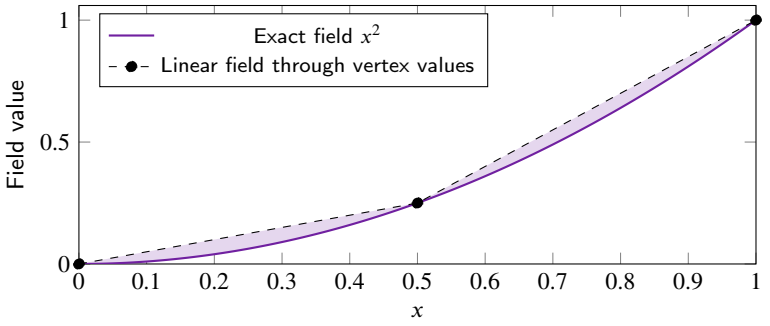


Figure 2: Exact vertex values can still give an inexact integral. With two segments ($h = 1/2$), the vertex values lie on x^2. Their median-dual weighted sum equals the area under the dashed piecewise-linear field: $Q_M = 3/8$, whereas the exact quadratic integral is $1/3$. The shaded difference is $1/24 = h^2/6$.

2.6. Corrections applied to the transferred state

The base transfer may create new extrema, change the solver's integral, or produce values incompatible with the boundary and gas-state constraints. The following steps address these different requirements; satisfying one does not guarantee the others.

2.6.1. Limiting new extrema

For the point methods, the Barth–Jespersen (BJ) limiter [20] provides a single factor $0 \le \theta_i \le 1$ at source vertex i. It reduces the linear reconstruction increment wherever the predicted values would exceed the range of the neighboring source values. The same factor multiplies the Taylor correction in NN2 and NN3 or the gradient correction in BAR3. Thus $\theta_i = 1$ retains the full correction and $\theta_i = 0$ gives NN1 or BAR2, respectively. The smooth-field verification uses $\theta_i = 1$. In the discontinuous tests labeled BJ, the factor is computed from the linear slope. For NN3 that check alone gives no bound on the added quadratic term, so extrema are measured on the final transferred field.

GP and CV use a different operation after solving for the target values. At each target vertex they use extrema from the overlapping source vertices. Hard limiting clips to those bounds. With relaxation, the lower and upper bounds are widened by τ times the source range, where τ is dimensionless. The correction introduced by clipping is redistributed to preserve the simplicial lumped total for GP or Q_B for CV. This step can change Galerkin orthogonality or individual CV integrals.

CV2 denotes the limited transfer, and CV2-u the unlimited construction. Flow comparisons use the same bound-relaxation tolerance as GP2, $\tau = 2 \times 10^{-3}$. Limiting, boundary conditions, and positivity corrections can change the unlimited field. Its conservation properties must therefore be checked again in the field actually used by the solver.

2.6.2. Bounded global correction

Bounded global correction (GC) can enforce Eq. (1) for any base transfer. For an uncorrected target field ϕ_B^{pre}, the nodal correction vector $\chi = (\chi_1, \dots, \chi_{n_B})^T$ satisfies

$$\sum_j w_j^B \chi_j = Q_A(\phi_A) - Q_B(\phi_B^{\mathrm{pre}}), \qquad \phi_j^{\min} \le \phi_{B,j}^{\mathrm{pre}} + \chi_j \le \phi_j^{\max}. \tag{18}$$

Here $\phi_j^{\min}$ and $\phi_j^{\max}$ are the lower and upper bounds at target vertex j. The input to global correction is ϕ_B^{pre}, and its corrected output is $\phi_B^{\mathrm{GC}} = \phi_B^{\mathrm{pre}} + \chi$. The correction is distributed among vertices that can change without exceeding those bounds; vertices already at a bound remain fixed [21, 22]. The requested integral must be feasible within the weighted bounds. A small total integral error does not require every nodal correction to be small.

2.6.3. Boundary conditions

Boundary enforcement supplies the values required on the physical boundary after transfer. Boundary values are interpolated from the source solution on the matching physical boundary. This replaces the unconstrained transfer at those vertices and can change the local GP or CV identities. Coincident source and target wall vertices retain the source

wall condition. If the discretized surface changes, its new wall orientation can instead introduce a mismatch. The flow tests retain all boundary vertices, excluding this geometric effect.

2.6.4. Positive density and pressure

For the ideal-gas states used here, pressure is $p = (\gamma - 1)[\rho E - |\rho \boldsymbol{u}|^2/(2\rho)]$, with heat-capacity ratio $\gamma > 1$. Positive density and internal energy are therefore coupled requirements: separate bounds on the conserved components do not guarantee positive pressure.

For GP, CV, and globally corrected transfers, changes from boundary treatment and scalar lower bounds are followed by integral-preserving redistribution. Any remaining states with nonpositive density or internal energy are repaired while preserving the affected patch totals. These steps preserve the selected global integral but can change local constraints. Global redistribution can also alter a uniform far-field region: any vertex with room within its relaxed bounds can receive part of the correction. A transfer can therefore preserve the domain integral while changing the fluxes through open boundaries during the next flow step.

2.6.5. Locally conservative flux correction

The Riemann tests include an alternative that redistributes corrections only between neighboring target volumes. This locally conservative (LC) flux correction replaces each method's usual limiting, global correction, and boundary-state recovery. Let $\tilde{b}_j$ be the source field integral over target control volume j, as in Eq. (14). The corrected nodal content is

$$w_j^B \phi_j^{\mathrm{LC}} = \tilde{b}_j + \sum_k \theta_{jk} \mathcal{F}_{jk}, \qquad \mathcal{F}_{jk} = -\mathcal{F}_{kj}, \quad \theta_{jk} = \theta_{kj} \in [0, 1]. \tag{19}$$

The pair flux $\mathcal{F}_{jk}$ is a correction to the content of volume j exchanged with neighboring volume k; the sum is over vertices sharing a target element. The symmetric pair factor θ_{jk} is the fraction of this exchange allowed between target volumes j and k, and ϕ_j^{LC} is the corrected nodal value. Applied in full, they recover the integral-corrected unlimited transfer. Each exchange adds to one volume what it removes from its neighbor, so the domain sum is unchanged even when limiting permits only part of a flux. Local conservation here allows neighboring volumes to exchange content; it does not require each nodal content to remain equal to $\tilde{b}_j$.

The starting low-order state, $\tilde{b}_j / w_j^B$, is bounded. Iterative flux-corrected transport [23] adds as much of the correction as the bounds allow, with the same relative relaxation, $\tau = 2 \times 10^{-3}$, as standard GP2 and CV2 in the flow tests. Each pass attempts to apply the remaining flux within the available bounds. The calculations use a fixed maximum of 12 passes to limit the work per transfer. Each accepted pair exchange conserves the total on that pass. Reaching the cap can leave part of the intended correction unapplied. The reported errors and bounds apply to this 12-pass setting. Only the Riemann calculations use LC; the airfoil calculations use the standard treatment above.

3. Effect of remeshing on multistep time integration

BDF2 uses two stored solutions. If either changes during transfer, the next time-step equation changes. We compare a transferred restart with a reference continuation on the same target mesh, so both use the same spatial discretization. The BDF2 coefficients then give the exact change in the residual. Predicting the resulting state or load change also requires the spatial operator and the accuracy of the nonlinear solve.

3.1. Stored states and the target-mesh reference

The finite-volume mass matrix is $M_M = \mathrm{diag}(w_i^M) \otimes I_{n_U}$, where I_{n_U} is the identity on the state components and $\otimes$ denotes the Kronecker product. Each vertex weight is repeated for its n_U components. This diagonal matrix differs from the coupled Galerkin mass matrix H_B. Between remeshing events, the semi-discrete equations are

$$M_M \dot{U}_M + F_M(U_M, t) = 0. \tag{20}$$

Here t is physical time, $\dot{U}_M = \mathrm{d}U_M/\mathrm{d}t$, and F_M is the assembled spatial residual, including fluxes, source terms, and boundary treatment. The state contains every variable used by the time integrator, including RANS turbulence variables. Boundary constraints must be applied consistently to the stored states and the equations. At constant time step, BDF2 on the target mesh is

$$\Phi_B(U_B^{n+1}; U_B^n, U_B^{n-1}) = \frac{M_B}{\Delta t}\left(\tfrac{3}{2}U_B^{n+1} - 2U_B^n + \tfrac{1}{2}U_B^{n-1}\right) + F_B(U_B^{n+1}, t^{n+1}) = 0. \tag{21}$$

Here n is the physical time-step index, t^n the corresponding time, and $\Delta t = t^n - t^{n-1} > 0$ the constant step size. During this physical step, the solver holds the stored states U_B^n and U_B^{n-1} fixed. Its inner iterations update an estimate of U_B^{n+1} to reduce the residual Φ_B. Each mesh is stationary during a time step. Continuously moving meshes would require compatible geometric histories and a discrete geometric conservation law [24, 25]; Eq. (21) does not cover that case.

Let U_A^k and U_B^k be the reference solutions computed on meshes A and B for the same problem at time t^k. A hat identifies the run that uses the transferred histories. At each time level k, define its state difference from the target-mesh reference by

$$\varepsilon^k = \widehat{U}_B^k - U_B^k. \tag{22}$$

At the two stored times $k = n-1, n$, the restarted run receives $\widehat{U}_B^k = \mathcal{T}_{AB}(U_A^k)$. The complete map $\mathcal{T}_{AB}$ includes the chosen raw transfer, limiting, boundary treatment, and admissibility corrections. It is therefore distinct from the raw operators $\mathcal{I}_{AB}^m$ in Sec. 2.2. At either stored level the discrepancy ε^k is measured against a solution computed independently on B. The discretization on each mesh also affects this comparison. To separate the contributions, imagine starting from an exact field sampled at the vertices of each mesh. Write these samples as $U_{A,\mathrm{ex}}^k$ and $U_{B,\mathrm{ex}}^k$. Adding and subtracting their contributions gives

$$\begin{aligned}\varepsilon^k = {} & \underbrace{\mathcal{T}_{AB}(U_{A,\mathrm{ex}}^k) - U_{B,\mathrm{ex}}^k}_{\text{transfer of exact samples}} \\ & + \underbrace{\mathcal{T}_{AB}(U_A^k) - \mathcal{T}_{AB}(U_{A,\mathrm{ex}}^k)}_{\text{effect of the source discretization}} - \underbrace{(U_B^k - U_{B,\mathrm{ex}}^k)}_{\text{target discretization error}}.\end{aligned} \tag{23}$$

The middle term passes the two source fields through the complete transfer separately. This makes the identity valid even when a limiter changes the transfer. The restart tests measure their combined effect, ε^k.

3.2. How errors in the stored states change the next step

The trial state $U_{B,\mathrm{trial}}^{n+1}$ is the current estimate of the flow at t^{n+1}. It starts as the initial guess for the inner solve. Each inner update gives a new estimate at the same physical time. To isolate the effect of transfer, take any one estimate and evaluate Eq. (21) twice with that field held fixed. Use the transferred histories in the first evaluation and the target-mesh reference histories in the second. Subtracting the two residuals then measures the effect of changing the histories alone.

Proposition 1 (Residual perturbation from transferred states). *Let Φ_B be the constant-step BDF2 residual of Eq. (21), and let the transferred histories differ from the target-mesh reference states by ε^n and ε^{n-1}, Eq. (22). Then, for any common trial value $U_{B,\mathrm{trial}}^{n+1}$,*

$$\begin{aligned}&\Phi_B(U_{B,\mathrm{trial}}^{n+1}; U_B^n + \varepsilon^n, U_B^{n-1} + \varepsilon^{n-1}) - \Phi_B(U_{B,\mathrm{trial}}^{n+1}; U_B^n, U_B^{n-1}) \\ &\qquad = -\frac{M_B}{\Delta t}\eta^n, \qquad \eta^n = 2\varepsilon^n - \tfrac{1}{2}\varepsilon^{n-1}.\end{aligned} \tag{24}$$

Proof. Let $\Delta\Phi_B$ denote the first residual minus the second in Eq. (24). Substituting the two sets of histories into Eq. (21) gives

$$\begin{aligned}\Delta\Phi_B &= \frac{M_B}{\Delta t}\Big[\big(\tfrac{3}{2}U_{B,\mathrm{trial}}^{n+1} - 2\widehat{U}_B^n + \tfrac{1}{2}\widehat{U}_B^{n-1}\big) - \big(\tfrac{3}{2}U_{B,\mathrm{trial}}^{n+1} - 2U_B^n + \tfrac{1}{2}U_B^{n-1}\big)\Big] \\ &\quad + F_B(U_{B,\mathrm{trial}}^{n+1}, t^{n+1}) - F_B(U_{B,\mathrm{trial}}^{n+1}, t^{n+1}) \\ &= \frac{M_B}{\Delta t}\big[-2(\widehat{U}_B^n - U_B^n) + \tfrac{1}{2}(\widehat{U}_B^{n-1} - U_B^{n-1})\big] = -\frac{M_B}{\Delta t}\eta^n.\end{aligned}$$

Both residuals contain the same trial value in the time derivative and the same spatial residual. These terms cancel exactly. □

Inserting the same error vector e in the latest state alone or in the older state alone gives, directly from Eq. (24),

$$\Delta\Phi_{\text{latest}} = -2M_B e/\Delta t, \qquad \Delta\Phi_{\text{older}} = M_B e/(2\Delta t) = -\tfrac{1}{4}\Delta\Phi_{\text{latest}}.$$

The factor $-1/4$ is thus a consequence of the BDF2 coefficients. The residual identity holds for finite history errors and even for nonlinear F_B, because both evaluations use the same trial value, time, mesh and spatial operator. The cancellation uses no Taylor expansion. At a boundary the same constraint must be applied to the prediction and the reported residual. For example, a row replaced by a prescribed value independent of the histories has zero history contribution. To obtain the corresponding change in the computed solution, we must also account for the spatial residual and the accuracy of the solve.

After the two restarts advance one step, Eq. (22) gives $\varepsilon^{n+1} = \widehat{U}_B^{n+1} - U_B^{n+1}$. Both states at this level are computed on B. The time index distinguishes this measured response from the input errors ε^{n-1} and ε^n. The remaining algebraic residuals are $\mathfrak{r}_B^{n+1} = \Phi_B(U_B^{n+1}; U_B^n, U_B^{n-1})$ and $\widehat{\mathfrak{r}}_B^{n+1} = \Phi_B(\widehat{U}_B^{n+1}; \widehat{U}_B^n, \widehat{U}_B^{n-1})$. These are the values of the left-hand side of Eq. (21) when each solve stops. Both vanish for exact solves. To compare the two computed states we subtract the residual of the reference run from that of the transferred run, using the same sign convention throughout. This gives

$$\begin{aligned} A_B\varepsilon^{n+1} &= M_B\eta^n + \Delta t\big(\widehat{\mathfrak{r}}_B^{n+1} - \mathfrak{r}_B^{n+1}\big) - \Delta t\,\mathcal{N}_B(\varepsilon^{n+1}), \\ A_B &= \tfrac{3}{2}M_B + \Delta t J_B, \qquad J_B = \left.\frac{\partial F_B}{\partial U}\right|_{U_B^{n+1},\,t^{n+1}}. \end{aligned} \tag{25}$$

The matrix A_B is the time-step Jacobian multiplied by Δt, and J_B is the spatial Jacobian. The nonlinear remainder is the change in the spatial residual left over after subtracting its linear prediction,

$$\mathcal{N}_B(\varepsilon^{n+1}) = F_B(\widehat{U}_B^{n+1}, t^{n+1}) - F_B(U_B^{n+1}, t^{n+1}) - J_B\varepsilon^{n+1}.$$

To bound it, join the reference state and the restarted state along a straight line in the space of discrete flow states. The states on this line are

$$(1-\sigma)U_B^{n+1} + \sigma\widehat{U}_B^{n+1} = U_B^{n+1} + \sigma\varepsilon^{n+1}, \qquad 0 \le \sigma \le 1.$$

At $\sigma = 0$ every vertex has the reference values. At $\sigma = 1$ every vertex has the restarted values. Intermediate σ interpolates each stored variable at the same vertex, with mesh and physical time held fixed.

Assume that along this segment the Jacobian change is at most L_F times the change in state, with finite constant L_F and the matrix norm induced by the chosen state norm. At fraction σ the distance from the reference is $\sigma\|\varepsilon^{n+1}\|$. Integrating the Jacobian along the segment therefore gives

$$\begin{aligned} \mathcal{N}_B(\varepsilon^{n+1}) &= \int_0^1 \big[\partial_U F_B(U_B^{n+1} + \sigma\varepsilon^{n+1}, t^{n+1}) - J_B\big]\varepsilon^{n+1}\,\mathrm{d}\sigma, \\ \|\mathcal{N}_B(\varepsilon^{n+1})\| &\le \int_0^1 L_F\sigma\|\varepsilon^{n+1}\|^2\,\mathrm{d}\sigma = \tfrac{1}{2}L_F\|\varepsilon^{n+1}\|^2. \end{aligned}$$

This local Lipschitz assumption on the Jacobian gives a quadratic remainder. If A_B is invertible, the two solves are exact on the same local solution branch, and η^n is small, Eq. (25) therefore yields

$$\varepsilon^{n+1} = R_B\eta^n + O(\Delta t\|\eta^n\|^2), \qquad R_B = A_B^{-1}M_B. \tag{26}$$

The response matrix R_B converts stored-state errors into a next-state change. This approximation is local to the chosen mesh, state, step, and norm. For example, a small change in the computed state can cause the flow solver to start clipping a reconstructed face value that was previously inside its bounds. Across that switch the spatial residual can have different derivatives. Likewise, stopping the two solves early can leave a residual difference large enough to matter in Eq. (25). Both effects concern the computed next states. The exact residual identity holds for any field common to the two evaluations, including the initial guess or the converged reference state.

3.3. Aerodynamic response

Let G_B be a scalar aerodynamic output evaluated on mesh B. Its change after one step is $\Delta G^{n+1} = G_B(\widehat{U}_B^{n+1}) - G_B(U_B^{n+1})$. At a fixed time on this mesh, assume that the gradient $\nabla_U G_B$ changes at most in proportion to a small change in the flow state near U_B^{n+1}. The remainder in a first-order load prediction is then bounded by a constant times $\|\varepsilon^{n+1}\|^2$.

Evaluating the gradient at U_B^{n+1} and substituting Eq. (26) into the first-order expansion gives

$$\Delta G^{n+1} = (\nabla_U G_B)^T R_B \eta^n + O(\|\eta^n\|^2). \tag{27}$$

For example, on a fixed airfoil in inviscid flow, the lift and drag coefficients are weighted sums of the wall pressure. Geometry, force direction and the reference quantities fix the surface weights. For an ideal gas with $\rho > 0$, pressure varies smoothly with the local conserved variables. In viscous flow, wall shear also contributes to the load. The derivative above compares nearby states at a single time. If an arriving disturbance changes the wall pressure sharply in one physical step, the load can change sharply in that step as well.

The sensitivity of the load to each stored-state error can be written explicitly using a discrete adjoint [26]. For this output, define the vector λ_G by the transpose solve $A_B^T \lambda_G = \nabla_U G_B$. At target vertex j, write $\lambda_{G,j}$ and η_j^n for the entries of the adjoint and the history combination at that vertex. In two-dimensional Euler flow each is a four-entry vector, ordered as density, streamwise momentum, transverse momentum and total energy per unit volume. Since $R_B = A_B^{-1} M_B$, the leading term in Eq. (27) is

$$(\nabla_U G_B)^T R_B \eta^n = \lambda_G^T M_B \eta^n = \sum_{j=1}^{n_B} w_j^B \, \lambda_{G,j}^T \eta_j^n. \tag{28}$$

The dot product multiplies each history error by its own load sensitivity at that vertex and adds the products. For example, the density error is multiplied by the sensitivity to density. These adjoint sensitivities depend on the chosen load through its gradient and on the target flow through A_B.

Consider mass conservation at a stored time level k. Let ρ_A^k and ρ_B^k be the independently computed density fields and $\widehat{\rho}_B^k$ the density field transferred to B. If transfer preserves the source mass, $Q_B(\widehat{\rho}_B^k) = Q_A(\rho_A^k)$. Subtracting the target-reference mass from both sides gives

$$\sum_{j=1}^{n_B} w_j^B \left(\widehat{\rho}_{B,j}^k - \rho_{B,j}^k\right) = Q_A(\rho_A^k) - Q_B(\rho_B^k). \tag{29}$$

Here $w_j^B = |C_j^B|$ is the area of the target control volume around vertex j in two dimensions, or its volume in three dimensions. The left-hand side weights each density difference by this measure and sums over the target vertices. The right-hand side is the mass difference between the two reference solutions. The same identity holds for each momentum component or for total energy when its integral is preserved. Even if the two reference masses agree, adding a given mass at one vertex and removing it at another leaves their total unchanged. For the same history level, those two changes cancel in first-order lift only if the density sensitivities in Eq. (28) are equal at the two vertices.

Equation (28) concerns the load one step after transfer. On the following step, BDF2 uses the newly computed difference ε^{n+1} and the retained input error ε^n. Its history combination is now $2\varepsilon^{n+1} - \varepsilon^n/2$. A load can therefore keep changing after a single mesh replacement. Section 4.4.2 tests this by disturbing a band away from the wall and measuring when pressure and lift respond.

4. Numerical tests and results

NN3, BAR3, and GP3 test third-order accuracy in smooth fields and its relation to bounds and integral preservation in discontinuous fields and the Riemann problem. Third-order variants are also included in the controlled BDF2 restarts. The repeated-remeshing Euler airfoil and RANS comparisons use NN2, GP2, and CV2 at a common nominal second order, which is the order of the CV formulation studied here. The Riemann comparisons use three random mesh-perturbation realizations per amplitude; the repeated-remeshing airfoil comparisons use two, and the RANS sensitivity tests use one. Local flux correction is tested in the Riemann problem.

4.1. Single-transfer tests

4.1.1. Smooth-field accuracy

A static test measures transfer alone, without evolving the flow. We first transfer the smooth asymmetric field

$$\phi(x, y) = \rho_\infty \left[2 + 0.3e^{0.8x/L_x + 0.5y/L_y} \sin(2\pi x/L_x) \sin(2\pi y/L_y)\right]. \tag{30}$$

Here x, y are Cartesian coordinates, ρ_∞ is reference density, and L_x, L_y are coordinate length scales. Target meshes are either structured quadrilaterals or independently generated mixed triangles and quadrilaterals, with nominal resolutions 64, 128, and 256. Source resolution is half the target resolution.

In this test, the analytic density is the manufactured field $\rho(x, y) = \phi(x, y)$. At target refinement level k, the relative nodal error is $e^{\text{node}}_{\rho,2,k} = [\sum_j (\rho_{B,j} - \rho(\boldsymbol{x}^B_j))^2 / \sum_j \rho(\boldsymbol{x}^B_j)^2]^{1/2}$. Equal vertex weights make this a point-value measure, rather than a volume-weighted field error. The dimensionless effective spacing is $h_{\text{eff},k} = n_{B,k}^{-1/2}$, where $n_{B,k}$ is that level's target vertex count. The observed order between successive levels is

$$p^{\text{obs}}_{k,k+1} = \frac{\log(e^{\text{node}}_{\rho,2,k} / e^{\text{node}}_{\rho,2,k+1})}{\log(h_{\text{eff},k} / h_{\text{eff},k+1})}. \tag{31}$$

The level index k is omitted when the mesh is clear from context. A least-squares fit of $\log e^{\text{node}}_{\rho,2}$ against $\log h_{\text{eff}}$ also gives an order over all three levels.

The NN, BAR, and GP methods recover their nominal smooth-field orders (Figure 3). NN1 is first order; NN2, BAR2, and GP2 are consistent with second order; NN3 is consistent with third order. BAR3 and GP3 have slopes slightly above three, approaching approximately 3.2 on the finest pair. On the finest mixed mesh, the relative errors are approximately 2.6×10^{-7} for BAR3, 3.4×10^{-7} for GP3, and 2.6×10^{-6} for NN3.

CV2 depends more strongly on element geometry. Its fitted order is 2.03 on the structured mesh, but 1.49 on the mixed mesh and 1.35 on its finest pair. This agrees with the loss of affine reproduction on quadrilaterals that remain distorted under refinement (Sec. 2.4). The relative mass-change magnitude $|e_Q|$ for CV2 is below 4×10^{-15} on both mesh families. The finest-mesh GP2 values of $|e_Q|$ are 4.3×10^{-12} (structured) and 2.5×10^{-7} (mixed).

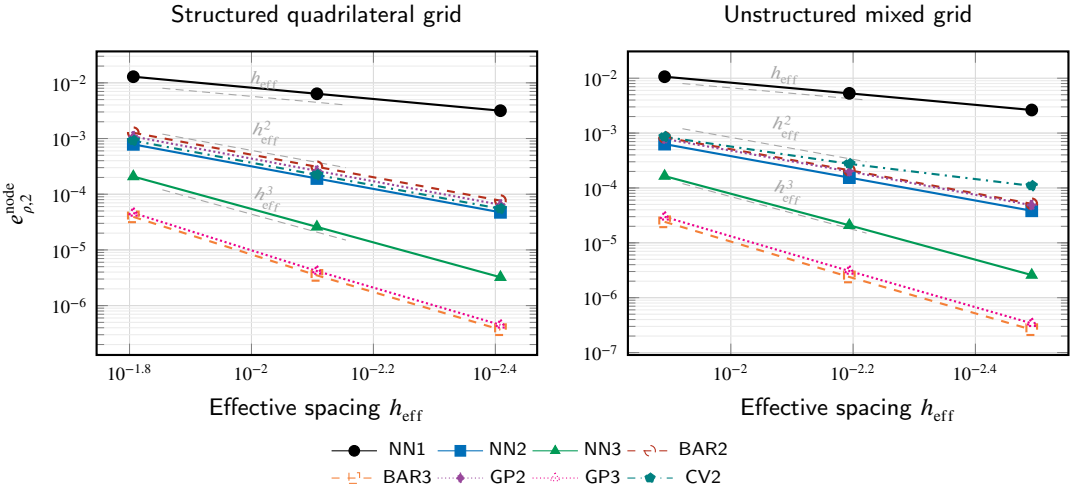


Figure 3: Smooth-field transfer accuracy on structured quadrilateral (left) and unstructured mixed (right) target meshes. The vertical axis shows the relative nodal density error $e^{\text{node}}_{\rho,2}$; h_{eff} is effective spacing derived from vertex count. NN, BAR, GP, and CV denote nearest-neighbor Taylor interpolation, barycentric interpolation, Galerkin projection, and control-volume transfer, with suffixes giving nominal accuracy order. Gray lines show first-, second-, and third-order slopes for comparison; they are not fits to the data.

4.1.2. Bounds and integral preservation at a discontinuity

The second field has a circular inclusion of radius $0.25\min(L_x, L_y)$, whose boundary crosses the mesh. Density is 1.2 outside and 3.6 inside. With temperature T, the full state is $T = 273(2+\sigma)$ K, $\rho = 1.2(2+\sigma)$, and $(u, v) = (100, 50)\sigma$ m/s, with $\sigma = 1$ inside and -1 outside. Momentum and total energy follow from these primitive variables using the gas constant $R_g = 287.05$ J/(kg K) and $\gamma = 1.4$. Density, temperature, and velocity in this static state are in kg/m^3, K, and m/s. Each conserved field is transferred independently. We compare unlimited and limited transfers by their density excursions beyond the source range and by the relative density-integral change $e_Q(\rho_A, \rho_B)$.

Preserving mass does not guarantee a bounded or physically admissible solution (Table 1, structured transfer $64 \to 128$). Unlimited CV2 preserves the density integral to roundoff but produces values outside the source range. Unlimited NN2 gives nonpositive internal energy or pressure near the interface and is rejected.

Limiting removes the measured new extrema, but can change the integral. In the BJ-limited NN2 test the reconstruction is suppressed at the jump, and the relative mass change reaches the NN1 value. Applying GC afterward restores the mass (Table 1). No new density extrema appear after that correction. Hard-limited GP2 and CV2 satisfy both density bounds and near-roundoff integral preservation in this test. GP3 remains bounded after restriction to the solver vertices, but its finite-volume integral has a nonzero error, consistent with the representation difference in Eq. (17).

Table 1
Representative discontinuous-field transfers. Overshoot and undershoot measure excursions beyond the exact density range $[1.2, 3.6]$; the last column gives the signed relative mass change $e_Q(\rho_A, \rho_B)$. BJ denotes Barth–Jespersen limiting and GC the bounded global conservation correction. Unlimited NN2 is omitted because it produces a physically inadmissible state.

Transfer	Overshoot	Undershoot	Relative mass change
NN3, unlimited	0.4259	0.2251	-5.52×10^{-7}
NN2, BJ	0	0	-4.46×10^{-3}
NN2, BJ+GC	0	0	2.99×10^{-14}
BAR3, unlimited	0.1766	0.1811	-8.80×10^{-7}
BAR3, BJ+GC	0	0	5.34×10^{-14}
GP2, unlimited	0.1346	0.1357	-2.52×10^{-15}
GP2, hard limited	0	0	1.06×10^{-15}
GP3, hard limited, vertex output	0	0	-4.83×10^{-7}
CV2, unlimited	0.0730	0.1807	5.83×10^{-15}
CV2, hard limited	0	0	6.62×10^{-15}

In a separate smooth transfer, global correction reduces the relative density-integral change to order 10^{-15} and changes the density L_1 error by at most 0.8%. Restoring the integral has little effect on point-value accuracy in that single transfer.

4.2. Repeated remeshing of a Riemann problem

4.2.1. Setup and error measures

The Riemann problem tests whether the behavior of one transfer predicts what happens after many transfers. We solve the two-dimensional Euler equations on $[0, 5] \times [0, 1]$, starting from a uniform reference mesh with 501×101 vertices and spacing $h = 0.01$. The nondimensional gas parameters are $\gamma = 1.4$ and $R_g = 1$. Initially,

$$(\rho, u, v, p) = \begin{cases} (1, 0, 0, 1), & x < 2, \\ (0.125, 0, 0, 0.1), & x \geq 2, \end{cases} \tag{32}$$

giving left and right temperatures of 1 and 0.8. Upper and lower boundaries are inviscid slip walls. Characteristic conditions at $x = 0$ and $x = 5$ prescribe total pressure and temperature $(p_0, T_0) = (1, 1)$ and $(0.1, 0.8)$, respectively, with direction $(1, 0)$ at both ends.

The calculation runs for 400 physical steps with $\Delta t = 10^{-3}$. At each remeshing event, interior vertices receive fresh random streamwise displacements relative to the original mesh. Boundary vertices remain fixed, and geometric distortion does not accumulate from mesh to mesh. If $a_{i,\min}$ is the smallest altitude of the corner triangles incident to vertex i, its maximum displacement is $a_{i,\min}/s$. We use deformation parameters $s = 4$ and $s = 2$; the latter allows twice the displacement.

Remeshing every ten or two physical steps compares less and more frequent transfer. Both BDF2 states are transferred at every mesh change. The lower-frequency comparison has 80 transfer evaluations; the higher-frequency integral record has 398. A fixed-mesh calculation with the same initial and boundary conditions supplies a numerical reference. Differences from it include both transfer error and the changed spatial discretization.

The density profile is averaged in y. A first width measure uses the window $W_1 = [2.40, 3.00]$ and estimates plateau densities at its ends. Position is the 50% crossing; width is the distance between the 10% and 90% crossings. Their differences from the fixed-mesh reference, Δx and Δw^{wide}, are expressed in cells of spacing $h = 0.01$. After many transfers, the smeared contact can enter the left plateau interval [2.40, 2.46]. The measured width then combines contact and shock distortion. A second, shock-only measure uses [2.55, 2.85] to exclude the contact tail; its width change is $\Delta w^{\mathrm{shock}}$. The wide-window tables use 16 sampling lines and 800 probes; the shock-only measure uses the stored eight-line mean profile at the same streamwise resolution. Both are evaluated at step 399 against their corresponding fixed-mesh references.

We measure mass changes in two ways. First, the change immediately across a transfer is ΔI_ρ, target minus source, with $I_\rho = Q_M(\rho_M)$. The normalized signed sum $\sum \Delta I_\rho / I_{\rho,0}$ includes transfer alone. Its common denominator is the initial mass, so it is distinct from a sum of the per-transfer changes e_Q. Second, final mass drift is $[I_\rho(t) - I_{\rho,0}]/I_{\rho,0}$, which also includes changes during flow evolution. Here $I_{\rho,0}$ is initial mass. The exact mass and energy for this Sod problem are 2.375 and 5.75, since no physical wave reaches an open boundary by $t = 0.399$. Let $e^{\mathrm{FV}}_{\rho,1}$ denote the median-dual-volume-weighted integral of the absolute density error against the exact Sod solution [27], divided by the domain volume. In the LC comparison, subscripts std and LC identify this same error with standard transfer and local flux correction.

The LC comparison contains 36 matched pairs covering NN1–NN3, BAR2–BAR3, GP2–GP3, and CV2. All methods use intervals of ten and two steps at $s = 4$. With the shorter interval, NN1, BAR2, GP2, and CV2 also use three realizations at both $s = 4$ and $s = 2$. Each standard/LC pair follows the same mesh sequence and ends on the same mesh. NN1+GC has no LC counterpart. Percentage changes in density error are computed within each pair before averaging. Final mass and energy drift are measured from the double-precision restart after the last transfer.

4.2.2. Contact smearing and shock broadening

Repeated transfer can smear the contact as well as the shock (Figure 4). After 398 transfers, the NN1 and BAR2 contact near $x \simeq 2.370$ spreads into the wide-window left plateau interval. Its estimated density can increase from approximately 0.266 to 0.287, moving the 10% crossing into the contact tail. Apparent broadening then exceeds 20 cells in the most diffused cases, but this is not a shock thickness. The shock-only measure moves the plateau sampling closer to the shock to exclude the contact.

The shock-only measure gives mean broadening after 398 transfers of 1.43 cells for NN1 at $s = 4$ and 8.35 at $s = 2$. BAR2 gives 1.07 and 2.21 cells. For GP2 and CV2 the mean change in shock width over the three realizations is below 0.02 cell in magnitude at both deformation amplitudes. Their wide-window changes are also small, at most 0.07 cell in magnitude. GP2 and CV2 thus preserve the tested wave profiles more closely than NN1 and BAR2, even after contact smearing is excluded from the shock measurement.

The mass integral tells a different part of the story. NN1+GC keeps the normalized signed sum of transfer mass changes below 1.0×10^{-13} in all realizations, yet its wave profile remains strongly smeared. CV2 also keeps this transfer-only sum near roundoff, while GP2 gives values of order 10^{-6}–10^{-5}. Preserving mass during transfer therefore does not preserve its spatial distribution. It also says nothing by itself about mass changes during the intervening flow steps.

4.2.3. Mass balance and local flux correction

Final mass drift reveals changes that the transfer-only sum misses (Figure 5a). Standard CV2 gives final relative mass drift of approximately 3.0×10^{-7}–4.1×10^{-7} at $s = 4$ and 5.5×10^{-6}–6.4×10^{-6} at $s = 2$, despite preserving mass during transfer to near roundoff. GP2 also has finite final drift, reaching about 2×10^{-6} at $s = 4$. The final-step budgets place the CV2 change in the flow step: at $s = 2$, realization 1, step 399 changes mass by 2.7×10^{-8} of the initial mass, and the following transfer change is negligible. With LC the recorded flow-step and transfer mass changes for the same case and step are both zero at the reported precision.

The standard GP/CV limiter clips values to its bounds and redistributes the integral change globally, according to the remaining capacity at each vertex. With bound relaxation 2×10^{-3}, even a constant far-field region can receive a correction. Measured density disturbances reach 1.4×10^{-4} for $x < 1$, and momentum density at the open boundaries

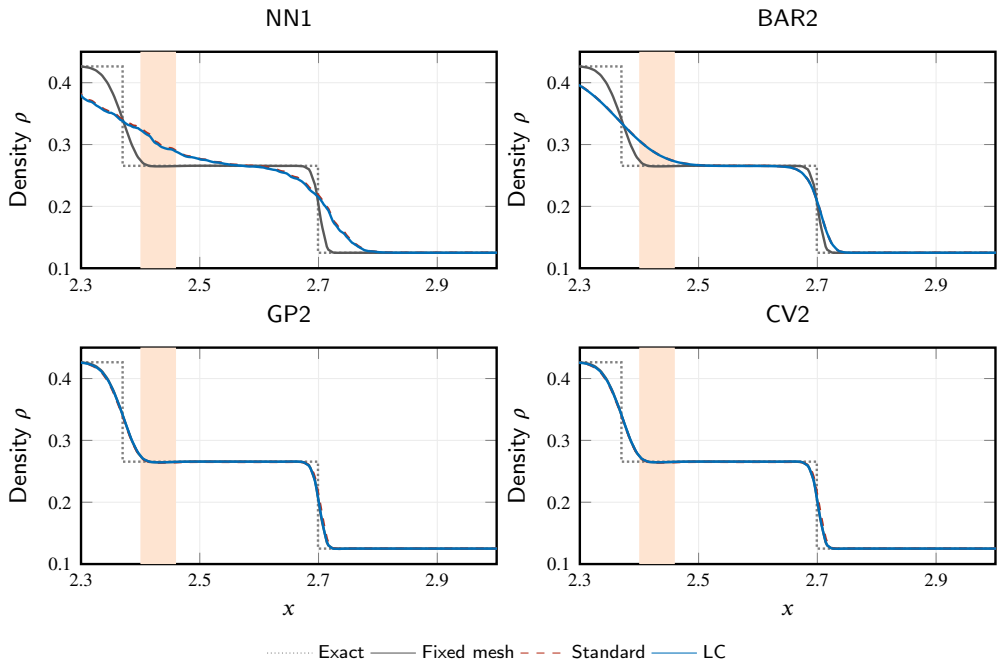


Figure 4: Density profiles after 398 transfers at $t = 0.399$, with remeshing every two steps, $s = 2$, and deformation realization 1. Each numerical curve is an eight-line mean. The exact Sod solution identifies the contact at $x \simeq 2.370$ and shock at $x \simeq 2.699$. Shading marks the left plateau interval $[2.40, 2.46]$ used by the wide-window diagnostic: contact diffusion raises this plateau for NN1 and BAR2. Shock-only widths instead use $[2.55, 2.85]$. Locally conservative flux correction (LC) has little effect on the visible broadening; several standard and LC curves overlap.

reaches approximately 2×10^{-4}. NN and BAR retain the constant far-field states to roundoff. The measurements connect global redistribution with far-field disturbances and later boundary mass exchange. A separate test with the limiter disabled was not performed, so its effect has not been isolated in that way.

LC keeps final relative mass drift below 5×10^{-13} and energy drift below 7×10^{-13} in all 36 matched cases. Its effect on density error is smaller (Figure 5b). Mean changes relative to standard transfer are -3.3% for GP and -2.9% for CV, compared with $+0.2\%$ for NN and -0.3% for BAR. At 398 transfers and $s = 2$, GP2 errors decrease by 7.0–7.9% and CV2 errors by 5.1–5.4% across the three realizations. Their wide-window position errors fall from +0.17–+0.20 cell to 0.00–+0.02 cell. For CV2 at $s = 2$, for example, the mean shock-only broadening is 0.011 cell with standard transfer and 0.029 cell with LC. Local flux correction therefore improves mass balance and some solution errors without removing the diffusion of the underlying transfer.

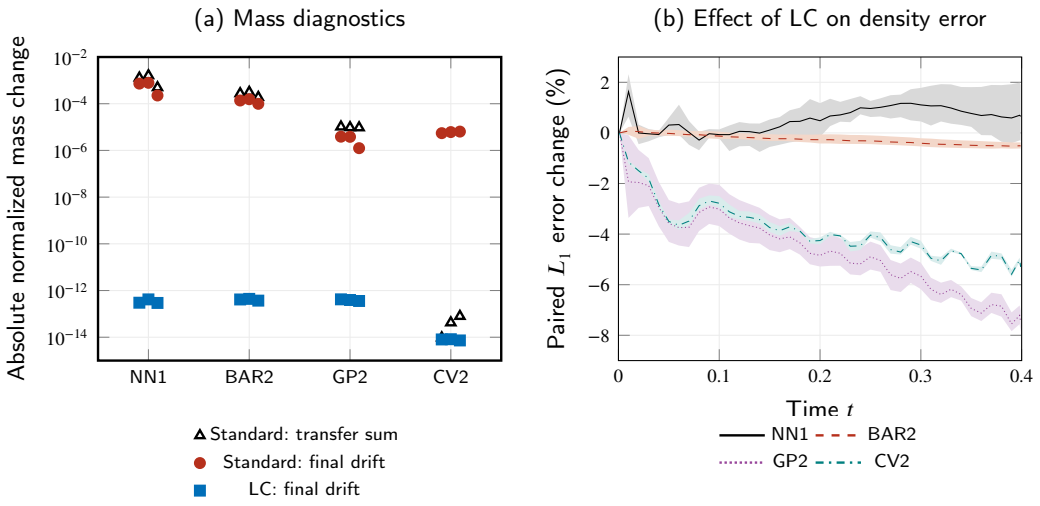


Figure 5: Mass balance and density accuracy for remeshing every two steps with $s = 2$. (a) Each symbol is one of three deformation realizations, offset horizontally for visibility. The absolute signed sum of standard transfer mass changes excludes CFD evolution; final drift uses the direct mass integral after the last transfer. Both are normalized by initial mass. (b) Paired changes $100(e^{\mathrm{FV}}_{\rho,1,\mathrm{LC}}/e^{\mathrm{FV}}_{\rho,1,\mathrm{std}} - 1)$, with density L_1 error measured against the exact Sod solution. Lines are means and shaded bands span the three realizations, not confidence intervals. Samples precede transfer, except the final sample at $t = 0.399$, which follows the last transfer. Negative values indicate lower error with LC.

The relaxed bounds allow small new extrema. Standard NN1 and BAR2 transfers are strictly monotone in this Riemann test, whereas LC introduces density excursions within the 2×10^{-3} relaxation. BAR2 excursions reach approximately 5.6×10^{-5} of the source range; NN2, NN3, and BAR3 excursions are limited to 2×10^{-3}. LC reaches its 12-pass limit in 23 of the 36 final transfers, so part of the unlimited correction remains unapplied. The fields remain conservative within the stated tolerance and satisfy the relaxed bounds, but need not recover the unlimited result. No admissibility repairs are activated. At slip walls, LC can introduce wall-normal momentum up to 1.3×10^{-4}, which the solver removes in the next step. Conservation and the relaxed bounds do not guarantee the exact wall condition immediately after transfer.

4.3. Controlled BDF2 restarts over an airfoil

4.3.1. Setup and error measures

Controlled airfoil restarts separate the effects of the latest and older stored BDF2 states. The Euler flow is over the National Advisory Committee for Aeronautics (NACA) 0012 airfoil at freestream Mach number $M_\infty = U_\infty/a_\infty = 0.1$ and angle of attack $\alpha = 6°$, with $p_\infty = \rho_\infty = T_\infty = 1$, $a_\infty = \sqrt{1.4}$, $U_\infty \simeq 0.118322$, and $\Delta t = 10^{-3}$. The subscript ∞ denotes freestream values, a_∞ is sound speed, U_∞ is flow speed, and c is airfoil chord. These Euler quantities are nondimensional, with unit reference chord $c = 1$.

The source and independent target meshes have 16,255 and 22,508 vertices, with 32,077 and 44,583 triangles. They share 370 airfoil vertices and a 63-vertex farfield polygon at $100c$, keeping the discretized boundary unchanged. Forces use pressure only, unit reference chord and area, and a moment reference at $x/c = 0.25$. Lift and drag coefficients are C_L and C_D. Any nonzero steady drag is numerical, rather than skin friction or shock drag.

Both meshes start from independently computed steady solutions, denoted by the superscript st. Each steady solution supplies two identical history levels. For each transfer method we form a single error field on the target mesh, $\varepsilon^{\mathrm{st}} = \mathcal{T}_{AB}(U^{\mathrm{st}}_A) - U^{\mathrm{st}}_B$. We add this same field to the latest history, the older history, both histories, or neither (Table 2). Each run then advances one step on B. Comparing it with the run that retains both reference histories isolates the contribution of the chosen history levels.

Let the subscripts latest, older and both identify these three perturbed runs. Their first-step state errors are all measured at t^{n+1}. According to Eq. (26), applying the same linear response matrix to the last column of Table 2 gives

$$\varepsilon^{n+1}_{\mathrm{latest}} \simeq \frac{4}{3}\varepsilon^{n+1}_{\mathrm{both}}, \qquad \varepsilon^{n+1}_{\mathrm{older}} \simeq -\frac{1}{3}\varepsilon^{n+1}_{\mathrm{both}}, \qquad \varepsilon^{n+1}_{\mathrm{both}} \simeq \varepsilon^{n+1}_{\mathrm{latest}} + \varepsilon^{n+1}_{\mathrm{older}}. \tag{33}$$

Table 2
Four restarts for each transfer method. Entries give the error added to each history and the resulting BDF2 history combination η^n.

Configuration	ε^n	ε^{n-1}	η^n
Reference	0	0	0
Both states	$\varepsilon^{\mathrm{st}}$	$\varepsilon^{\mathrm{st}}$	$3\varepsilon^{\mathrm{st}}/2$
Latest only	$\varepsilon^{\mathrm{st}}$	0	$2\varepsilon^{\mathrm{st}}$
Older only	0	$\varepsilon^{\mathrm{st}}$	$-\varepsilon^{\mathrm{st}}/2$

The comparison tests linear response to this particular error field. For the mean-flow variables, we measure the first-step difference using the volume-weighted norm $\|\varepsilon^{n+1}\|_{M_B} = [(\varepsilon^{n+1})^T M_B \varepsilon^{n+1}]^{1/2}$ and the Euclidean norm $\|\varepsilon^{n+1}\|_2 = [(\varepsilon^{n+1})^T \varepsilon^{n+1}]^{1/2}$. The first norm weights each vertex by its control-volume size. The second gives all vertices equal weight.

For each configuration, we compute the first-step lift coefficient from the wall pressure and subtract the corresponding value from the reference restart. This gives $\Delta C_{L,\mathrm{case}}$. To assess linear superposition, define the state and lift remainders and their relative errors by

$$\begin{aligned} \mathcal{E}_U &= \varepsilon^{n+1}_{\mathrm{both}} - \varepsilon^{n+1}_{\mathrm{latest}} - \varepsilon^{n+1}_{\mathrm{older}}, \\ \mathcal{E}_{C_L} &= \Delta C_{L,\mathrm{both}} - \Delta C_{L,\mathrm{latest}} - \Delta C_{L,\mathrm{older}}, \\ e^{U}_{\mathrm{sup}} &= \frac{\|\mathcal{E}_U\|_{M_B}}{\|\varepsilon^{n+1}_{\mathrm{both}}\|_{M_B}}, \qquad e^{C_L}_{\mathrm{sup}} = \frac{|\mathcal{E}_{C_L}|}{|\Delta C_{L,\mathrm{both}}|}. \end{aligned} \tag{34}$$

All reported denominators are nonzero. State-norm ratios are nonnegative; lift ratios retain their signs.

The solver reports the negative of the left-hand side of Eq. (21). For each pair of runs, we compare the residuals recorded before the first inner update, when both runs use the same initial guess for U_B^{n+1}. The older-only and reference runs both start from the steady target field U_B^{st}. Subtracting the reference history combination from the older-only one gives $\Delta\eta = -\varepsilon^{\mathrm{st}}/2$. The latest-only and both-states runs both start from the transferred steady field $\widehat{U}_B^{\mathrm{st}} = \mathcal{T}_{AB}(U_A^{\mathrm{st}})$. Subtracting the both-states history combination from the latest-only one gives $\Delta\eta = +\varepsilon^{\mathrm{st}}/2$. The predicted printed difference is $+M_B\Delta\eta/\Delta t$. Double-precision residuals are split into wall-normal and tangential components.

4.3.2. Effects of the latest and older stored states

Perturbing only the older state changes next-step lift even though the latest state matches the reference. For every method in Figure 6, its sign is opposite to that obtained by perturbing both states. GP2 gives lift differences of $+2.453 \times 10^{-3}$ and -7.315×10^{-3}, respectively; CV2 gives $+2.439 \times 10^{-3}$ and -7.272×10^{-3}.

The measured residual changes agree with the exact BDF2 identity to relative errors of approximately 10^{-7} for all tested transfers. This also holds at the 370 slip-wall vertices. Their source and target positions coincide, keeping the transferred states compatible with the target slip condition. The wall-normal momentum mismatch between the measured and predicted residual changes is at most 4.5×10^{-12} for the four transfers checked at the wall. The tangential mismatch is $(1.1\text{–}1.3) \times 10^{-7}$ of the tangential perturbation.

Restoring the global integral barely changes first-step lift. With both states perturbed, NN1 changes from 5.7873×10^{-3} to 5.7961×10^{-3}, BAR2 from -7.2852×10^{-3} to -7.2849×10^{-3}, and CV2 from -7.2724×10^{-3} to -7.2765×10^{-3}. CV2 already preserves the source density integral before this correction. These forward calculations show how little the first-step lift changed when the integral was restored. Equation (28) explains why an integral correction alone cannot predict the lift change.

The second- and third-order methods give state-error norms near 7.0×10^{-4}, with less than 3% spread; NN1 gives 1.47×10^{-3}. These are differences from the independently computed target-mesh reference. They include both transfer error and mesh-dependent steady-solution differences, so they do not rank interpolation accuracy alone.

Across all eight variants, the latest-only/both-states state-norm ratio is 1.3330–1.3333, and the older-only/both-states ratio is 0.3335–0.3341. The relative superposition errors are 1.8–3.5% in the volume-weighted state norm and 0.28–1.49% in lift. The ratios support the linearized prediction in Eq. (33) for this imposed error field. Additional cancellation and amplitude tests show where this approximation remains unverified.

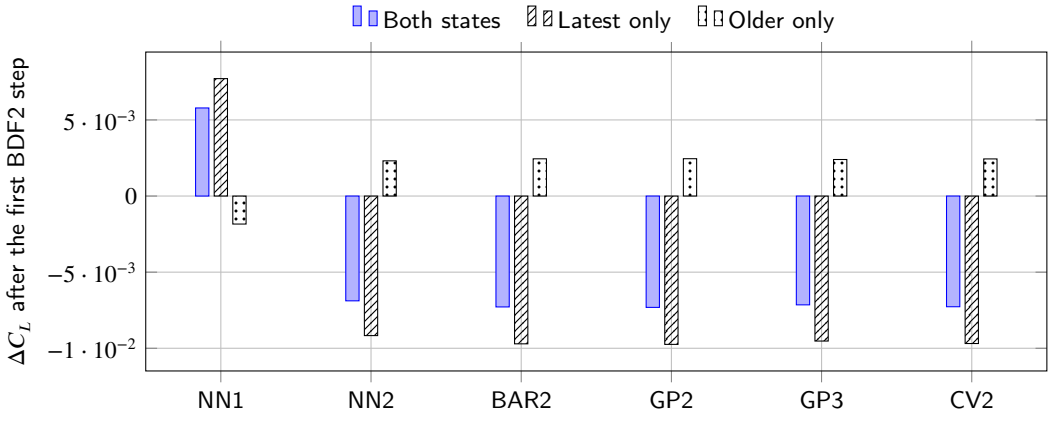


Figure 6: First-step lift-coefficient difference ΔC_L after transfer to the target Euler airfoil mesh, relative to the continuation of the target-mesh reference. NN, BAR, GP, and CV denote nearest-neighbor Taylor interpolation, barycentric interpolation, Galerkin projection, and control-volume transfer; suffixes indicate nominal accuracy order. For each method, the same error field is imposed in both stored states or in one state alone. The opposite signs of the responses with only the older state perturbed and with both states perturbed are consistent with the BDF2 coefficients.

4.3.3. Limits of the linear-response approximation

Setting $\epsilon^n = \epsilon^{\mathrm{st}}/4$ and $\epsilon^{n-1} = \epsilon^{\mathrm{st}}$ gives $\eta^n = 0$, so the reference and perturbed BDF2 equations are identical when evaluated with the same boundary treatment. An exact solve on the same local solution branch would give the same next state. In the computed cancellation test, lift differences are of order 10^{-9}, but volume-weighted state differences remain 4.3–9.0% of the imposed discrepancy. Iterative error has not been independently quantified, so the small lift difference does not verify cancellation of the state response. This limitation does not affect the exact residual identity or its separate common-trial check.

We repeat the restarts with the full, half and quarter amplitudes of the same imposed history error. The result depends on how the state difference is measured. Lift and Euclidean state-norm responses approach superposition more closely for BAR2, GP2, and CV2. In the volume-weighted norm, however, the absolute remainder decreases slowly and the relative remainder grows from about 3.4% to 8.7–9.2%. These tests do not establish a uniform quadratic remainder in every norm. Even with an exact first step, the cancellation would last only one step. The retained error $\epsilon^n = \epsilon^{\mathrm{st}}/4$ would give $2\epsilon^{n+1} - \epsilon^n/2 = -\epsilon^{\mathrm{st}}/8$ at the next step.

4.4. Airfoil loads after remeshing

The following Euler tests use the flow conditions of Sec. 4.3.1.

4.4.1. Initial load growth under repeated remeshing

To examine accumulation in the airfoil calculation, we perturb vertices at wall distances between $0.01c$ and $20c$, keeping boundary vertices fixed. NN2, GP2, and CV2 use matching random deformations. A 400-step interval contains 39 remeshing events followed by flow evolution. Load differences are measured against the corresponding fixed-mesh trajectory. Fitted slopes per 100 physical steps describe this window; they are neither long-time drift estimates nor estimates of independent-sample uncertainty.

GP2 and CV2 give slower initial drag growth than NN2 in both realizations. GP2 slopes are 6.31×10^{-6} and 5.89×10^{-6} per 100 physical steps; CV2 gives 5.92×10^{-6} and 5.82×10^{-6}, compared with 3.95×10^{-5} and 3.84×10^{-5} for NN2. The mean NN2 slope is approximately 6.4 times the GP2 mean and 6.6 times the CV2 mean. However, 400 steps span only about 0.047 convective and 0.47 acoustic chord times. The result describes early growth, and lift trends vary more between realizations.

4.4.2. Delayed response to a disturbance away from the wall

A separate test isolates one transfer in a narrow band away from the wall. The bands begin at distances $0.05c$, $0.10c$, and $0.20c$. GP2 is used in all three, and CV2 repeats the longer $0.20c$-band calculation. Fixed shells measure the root-mean-square (RMS) pressure difference from a common fixed-mesh reference. Coupled transfer and implicit time

integration can introduce small immediate changes outside the displaced band. We therefore measure arrival when the signal first reaches a specified fraction of its peak. For a shell at wall distance r inside a band beginning at r_b, the low-Mach acoustic travel estimate is $(r_b - r)/(a_\infty \Delta t) \simeq 845(r_b - r)/c$ physical steps in these nondimensional Euler calculations. Transfer already produces immediate changes up to approximately $0.08c$ inward of the displaced band; these shells are excluded from propagation fits.

Beyond the region changed immediately by transfer, the pressure response is delayed (Figure 7). For GP2, the pressure difference first reaches half its local peak 55 steps after remeshing at $r/c = 0.10$ and 130 steps after remeshing at $r/c = 0.005$. These crossings follow a changing pressure field through time. A sharp change in a load trace as wall pressure responds is compatible with a smooth pressure integral on the fixed airfoil surface. For the longer calculation with the band beginning at $0.20c$, estimated propagation speeds are 1.09–1.27 with GP2 and 1.10–1.30 with CV2, using 10–75% amplitude thresholds. Both ranges include the sound speed $a_\infty \simeq 1.183$. Their spread shows the threshold sensitivity of a single event, rather than variability across independent events.

A separate restart test also shows why the first load difference is insufficient. The NN1 lift difference peaks three steps after the first post-transfer sample, at 4.66 times its initial value. The sampled second- and third-order responses peak at the first step and then decay.

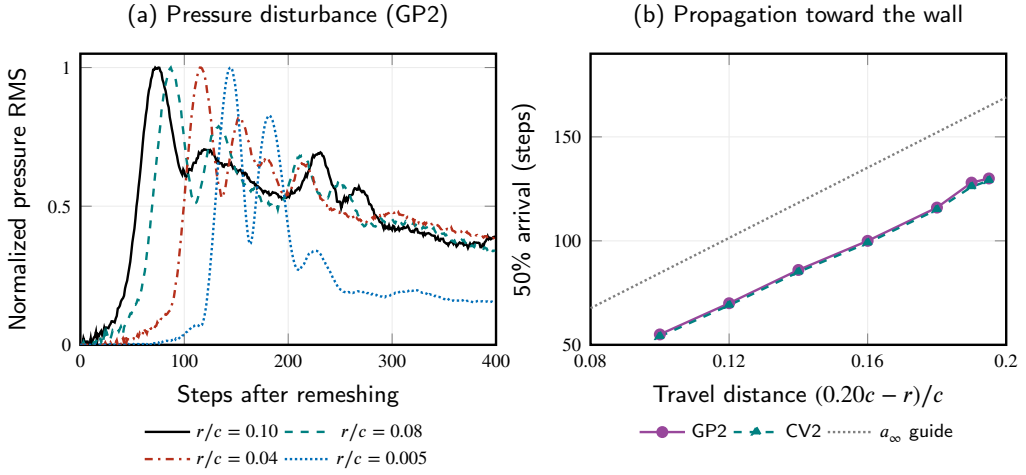


Figure 7: Pressure response to a single Euler remeshing event in the annulus $0.20c \le r \le 0.25c$, where r is distance from the airfoil wall and c is chord. (a) Shell RMS pressure differences from the fixed-mesh reference, normalized separately by each shell's maximum over the 400 post-event samples; shell half-width is $0.004c$. Disturbances reach shells closer to the wall later. (b) First crossings of 50% of each shell's maximum for GP2 and CV2, restricted to travel distances of at least $0.08c$. Connecting lines guide the eye; the dotted line gives travel time at the freestream sound speed. The points describe one event per method, not independent realizations.

4.5. Viscous airfoil restarts with the SA model

4.5.1. Setup and numerical sensitivity tests

For a viscous flow, does the transfer method that best preserves the integrals also keep drag closest to the fixed-mesh reference? We test this on the stationary airfoil with the one-equation SA model [28]. Each run starts from a steady RANS calculation. The prescribed mesh changes then disturb the flow. The conditions are $M_\infty = 0.1$, $\alpha = 6°$, and chord Reynolds number $Re_c = \rho_\infty U_\infty c/\mu_\infty = 6 \times 10^6$, where μ_∞ is freestream dynamic viscosity.

The coarse mesh has 23,271 vertices, 9,686 boundary-layer and wake quadrilaterals, 26,851 triangles, and 256 airfoil vertices. Medium and fine meshes have 32,102 and 48,228 vertices. All retain a first off-wall spacing of approximately $5 \times 10^{-6}c$. The farfield is at $100c$, and the quadrilateral region extends to about $0.11c$.

We use the Stanford University Unstructured (SU2) solver [29]. Mean-flow convection uses the Harten–Lax–van Leer contact (HLLC) approximate Riemann solver with a monotonic upstream-centered scheme for conservation laws (MUSCL) reconstruction. Turbulence transport uses scalar upwinding without MUSCL. BDF2 is solved by dual-time stepping at a pseudo-time Courant–Friedrichs–Lewy (CFL) number of 6. The nominal limit is 800 pseudo-time iterations

per physical step. After ten iterations, the solver stops if $\log_{10}\mathrm{rms}(r_\rho) \le -12$, with r_ρ the nodal density-equation residual in solver normalization and rms its root-mean-square over solver vertices.

The dimensional physical step $\Delta t_0 = 10^{-3}$ s corresponds to $\Delta t_0^* \simeq 0.347224$. A superscript $*$ denotes normalization by solver reference scales. With $U_\infty^* = 0.1$, the 400-step window spans approximately 13.9 chord flow-through times, measured by tU_∞/c. Startup uses a first-order step ten times smaller. The fixed-mesh SA reference advances for 800 steps from matching stored states.

The coarse steady SA solution reaches its 20,000-iteration limit at -11.2, with lift and drag stationary to 10^{-7} over the last 2,000 iterations. Table 5 reports convergence during the subsequent physical steps.

NN2, limited GP2, and limited CV2 follow the same prescribed mesh sequences. At each event, 12,602 vertices may move and 10,669 remain fixed. No vertex with solver wall distance below $0.01c$ moves. The geometric mask uses distance to the nearest airfoil vertex, allowing part of the outer quadrilateral region to move. The deformation parameter is $s = 4$; a minimum corner-area ratio of 0.2 maintains valid cells. Boundary vertices remain fixed. Remeshing occurs every ten physical steps, independently of the solution.

At each mesh change we transfer both stored time levels of the mean flow and the SA working variable $\tilde{\nu}$. We monitor the integral of $\rho\tilde{\nu}$ alongside the mass, momentum and total-energy integrals. Wall distance and eddy viscosity are recomputed. Integral changes use e_Q from Sec. 2.1; all monitored source integrals are nonzero. The SA equation contains source terms, so the integral of $\rho\tilde{\nu}$ can subsequently change during a flow step.

Pressure and viscous drag contributions are $C_{D,p}$ and $C_{D,v}$, with $C_D = C_{D,p} + C_{D,v}$. The label Q selects a monitored coefficient C_Q: lift, total drag, pressure drag, or viscous drag. For transfer method m, the remeshing response is the coefficient minus its fixed-mesh reference at the same numerical setting:

$$\Delta C_{Q,m}(t) = C_{Q,m}(t) - C_{Q,\mathrm{ref}}(t), \qquad P_Q(t) = \Delta C_{Q,\mathrm{NN2}}(t) - \Delta C_{Q,\mathrm{GP2}}(t). \tag{35}$$

Thus $\Delta C_{D,\mathrm{GP2}}$ measures how far the GP2 drag history departs from its fixed-mesh reference, and P_D measures the difference between the NN2 and GP2 drag histories on the same mesh sequence. Remeshed and reference calculations use identical surface quadrature. For any scalar history $q(t)$ sampled at N_s common physical times t_i, define $\mathcal{R}(q) = (N_s^{-1}\sum_{i=1}^{N_s} q(t_i)^2)^{1/2}$. Write q_{nom} and q_{test} for this history at the nominal and modified numerical settings. The sensitivity is

$$S_{\mathrm{test}}(q) = \frac{\mathcal{R}(q_{\mathrm{test}} - q_{\mathrm{nom}})}{\mathcal{R}(q_{\mathrm{nom}})}. \tag{36}$$

All reported histories have nonzero nominal RMS. Subtracting the two histories at each common sample time makes S_{test} sensitive to changes in both amplitude and timing.

For the first mesh sequence we compare 38 common time samples spanning three remeshing events. At Δt_0 we raise the inner-iteration limit from 800 to 1600. We also repeat the calculation at $\Delta t_0/2$ and $\Delta t_0/4$, each with the original 800-iteration limit. The same displacement fields are applied at matching physical times, and each step size has its own fixed-mesh reference. Medium and fine steady meshes assess absolute-load sensitivity. A short medium-mesh SA transient uses the same displacement statistics and amplitude limit, but a different nodal deformation field; its differences therefore combine mesh-resolution and disturbance changes.

4.5.2. Transferred integrals and aerodynamic loads

GP2 and CV2 change the transferred integrals less than NN2 (Table 3). GP2's maximum errors in the density and total-energy integrals are roughly two orders of magnitude smaller. For the integral of $\rho\tilde{\nu}$, the largest relative changes are 1.70×10^{-6} with NN2, 6.88×10^{-9} with GP2 and 1.99×10^{-12} with CV2. CV2 preserves each mean-flow integral to within a relative change of 1.2×10^{-14}.

Table 4 compares the three transfer methods over both mesh sequences. CV2 gives the smallest RMS drag departure from the fixed-mesh reference on the first sequence and the largest on the second. The last column gives the smaller difference between NN2 and GP2. We use this difference to test how strongly the physical time step affects the comparison.

In the 38-sample control window, halving the time step changes the GP2 drag response by 8.0% of its nominal RMS (Figure 8a). Over the same samples the NN2–GP2 drag difference has a nominal RMS of 2.19×10^{-6}. Halving the step changes this difference by 2.82×10^{-6} in RMS, or 129% of the signal being compared (Figure 8b). At the quarter step the change reaches 171% relative to the same nominal history. The corresponding changes in the lift difference are 75% and 132%.

Table 3
Largest absolute relative integral change across the repeated transfers of the stored states in the SA case. Each method column gives the maximum over both mesh sequences and both stored time levels. The last column is the NN2/GP2 ratio of the separate maxima.

Integrand	NN2	GP2	CV2	NN2/GP2
ρ	$8.41 \cdot 10^{-11}$	$4.56 \cdot 10^{-13}$	$7.19 \cdot 10^{-15}$	185
ρu	$3.78 \cdot 10^{-9}$	$5.76 \cdot 10^{-11}$	$9.62 \cdot 10^{-15}$	66
ρv	$1.65 \cdot 10^{-8}$	$8.08 \cdot 10^{-11}$	$1.08 \cdot 10^{-14}$	204
ρE	$9.54 \cdot 10^{-11}$	$5.82 \cdot 10^{-13}$	$1.19 \cdot 10^{-14}$	164
$\rho\tilde{\nu}$	$1.7 \cdot 10^{-6}$	$6.88 \cdot 10^{-9}$	$1.99 \cdot 10^{-12}$	247

Table 4
RMS drag differences for the SA case over 398 samples on each mesh sequence, at the nominal time step. The first three method columns compare each drag history with its fixed-mesh reference. The last column compares the NN2 and GP2 histories directly. Multiply each entry by 10^{-6} to obtain the drag-coefficient difference.

Mesh sequence	NN2	GP2	CV2	NN2 minus GP2
1	63.84	64.32	63.26	7.72
2	60.85	60.90	61.90	9.57

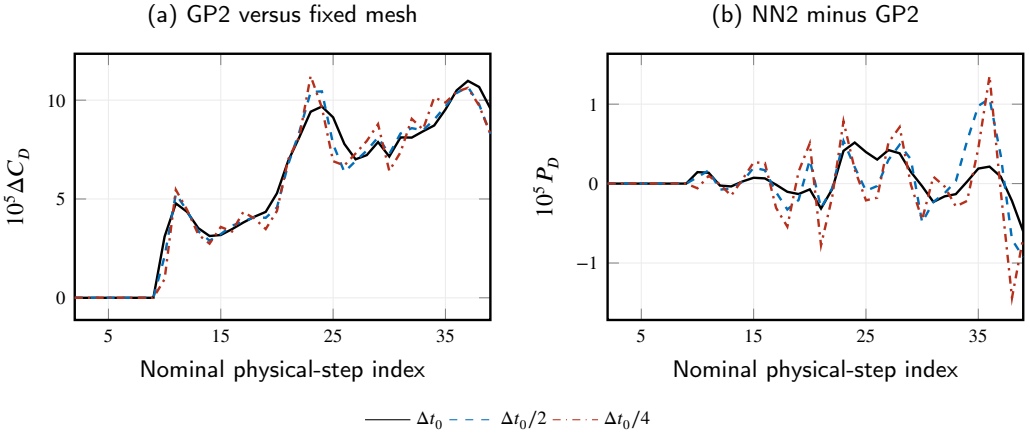


Figure 8: Time-step sensitivity of drag with SA over three remeshing events for deformation realization 1. Panel (a) shows the GP2 drag difference ΔC_D from the fixed-mesh reference at the same time step. Panel (b) shows the NN2–GP2 drag difference, $P_D = C_{D,\mathrm{NN2}} - C_{D,\mathrm{GP2}}$. All curves use common physical sample times; the horizontal axis gives time divided by the nominal step Δt_0. Mesh perturbations are applied at matching physical times.

4.5.3. Numerical convergence and limits of the comparison

Table 5 distinguishes steps that meet the residual criterion from those that stop at an iteration limit. On the first mesh sequence, the nominal runs reach the 800-iteration limit on 147–155 of 398 steps, depending on the transfer method. All three methods meet the density criterion on every step of the short 1600-iteration test and the half- and quarter-step tests.

When the iteration limit is doubled, the largest RMS change in an individual lift or total-drag history is 0.34% of its nominal RMS over the control window. The NN2–GP2 difference changes by 10.9% for lift and 8.4% for drag.

The drag components also require care. In the SA quarter-step test, friction-drag history differences are approximately 76–92% of the nominal RMS friction-drag response. The smaller sensitivity of total drag does not establish convergence of its separate components.

As the steady SA mesh is refined from 23 271 to 32 102 and 48 228 vertices, total drag C_D decreases from 0.010229 to 0.010048 and 0.009996. These are changes in the absolute load. The medium-mesh restart gives an RMS NN2–GP2 drag

Table 5
Inner-iteration convergence for SA on the first mesh sequence. $N_{\max}$ is the iteration limit. Separate columns give the number of steps reaching that limit for each method. Nominal rows cover the full interval; other rows cover the short sensitivity window.

			Steps at limit		
$N_{\max}$	$\Delta t/\Delta t_0$	Steps	GP2	NN2	CV2
800	1	398	152	155	147
1600	1	38	0	0	0
800	1/2	78	0	0	0
800	1/4	158	0	0	0

difference of 9.94×10^{-7} over the short comparison. Changing the mesh also changes the imposed vertex displacements, so this test cannot isolate the effect of resolution on that difference. The integral comparison favors GP2 and CV2. The drag comparison yields different orderings on the two mesh sequences, and the time-step changes exceed the small NN2–GP2 drag signal in the control window. An aerodynamic ranking for this SA case therefore remains unresolved.

4.6. Interpreting the comparisons

Spatial profiles, integrals, state norms, and loads measure different aspects of transfer error. Differences from a target-mesh reference also include differences between the independently computed mesh solutions. These distinctions agree with studies that separate integral preservation from other discrete properties [6], and preserve discrete equilibria to suppress transfer-induced disturbances [12, 13].

CV2 preserves the global finite-volume integral but loses affine reproduction on persistently distorted quadrilaterals. Limiting and boundary treatment can then change its local constraints. LC replaces global redistribution with exchanges between neighboring volumes. This treatment is tested only in the Riemann problem; the standard-transfer airfoil tests establish no LC aerodynamic benefit.

Computational assistance Generative artificial intelligence (AI) tools, ChatGPT (OpenAI) and Claude (Anthropic), assisted the design of verification tests, code development, and numerical post-processing. The tabulated values and data-driven figures are obtained from numerical outputs through the definitions given above, not from generative model predictions.

5. Conclusions

Transfer in an unsteady calculation must supply every state used by the time integrator. In the controlled BDF2 restarts, an error in the older state changes next-step lift even when the latest state matches the target-mesh reference. With the estimate of the new state held fixed, inserting the same error into the older history changes the residual by exactly $-1/4$ of the latest-history change. The measured lift changes also have opposite signs. Their magnitudes depend on how the target flow and the chosen load weight that error.

The Riemann results distinguish preservation of mass, wave shape, and mass balance during later flow evolution. Global mass correction does not restore a smeared profile. At high transfer counts, a wide-window 10–90% width also includes contact smearing, so it must be separated from shock-only broadening. GP2 and CV2 preserve the tested profiles more closely than NN1 and BAR2, but standard global limiting can disturb the far field and allow mass changes during subsequent flow steps.

Local flux correction keeps final mass and energy drift near roundoff and modestly reduces GP/CV density error without materially changing shock broadening. It allows small extrema within relaxed bounds and can leave part of the unlimited correction unapplied. Exact wall compatibility immediately after transfer is not guaranteed either.

In the SA airfoil restarts, CV2 produces the smallest changes in the monitored integrals. The ordering of the RMS drag departures depends on the mesh sequence, and halving the physical time step changes the NN2–GP2 drag difference by more than its nominal RMS. The viscous calculations therefore leave the aerodynamic ranking unresolved. Transfer accuracy must be assessed through both the transferred fields and the flow evolution that follows, including all stored time levels.

Data availability

The accompanying supplement collects the complete numerical tables and additional checks, including the companion viscous runs with the shear-stress-transport model [30]. The numerical inputs used to compile the figures and tables are available in the public data repository. The archive does not include every flow-field output from the simulations.

Funding

Funding information to be supplied.

Contributor Roles Taxonomy (CRediT) authorship contribution statement

Author contributions to be supplied.

Declaration of competing interest

Competing-interest declaration to be supplied.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

ChatGPT (OpenAI) and Claude (Anthropic) were used to assist literature synthesis, organization, scientific editing, and the preparation of the manuscript and data-driven figures.

References

[1] F. Alauzet, A. Loseille, G. Olivier, Time-accurate multi-scale anisotropic mesh adaptation for unsteady flows in CFD, J. Comput. Phys. 373 (2018) 28–63. doi:10.1016/j.jcp.2018.06.043.

[2] B. Sauvage, F. Alauzet, A. Dervieux, A space and time fixed point mesh adaptation method, J. Comput. Phys. 519 (2024) 113389. doi:10.1016/j.jcp.2024.113389.

[3] T. Levý, G. May, Towards a robust time-accurate anisotropically adaptive hybridized discontinuous Galerkin method, Comput. Fluids 301 (2025) 106792. doi:10.1016/j.compfluid.2025.106792.

[4] X. Jiao, M. T. Heath, Common-refinement-based data transfer between non-matching meshes in multiphysics simulations, Int. J. Numer. Methods Eng. 61 (2004) 2402–2427.

[5] P. E. Farrell, J. R. Maddison, Conservative interpolation between volume meshes by local Galerkin projection, Comput. Methods Appl. Mech. Eng. 200 (1–4) (2011) 89–100. doi:10.1016/j.cma.2010.07.015.

[6] J. R. Maddison, H. R. Hiester, Optimal constrained interpolation in mesh-adaptive finite element modeling, SIAM J. Sci. Comput. 39 (5) (2017) A2257–A2286. doi:10.1137/15M102054X.

[7] J. K. Dukowicz, J. W. Kodis, Accurate conservative remapping (rezoning) for arbitrary Lagrangian–Eulerian computations, SIAM J. Sci. Stat. Comput. 8 (3) (1987) 305–321. doi:10.1137/0908037.

[8] L. G. Margolin, M. Shashkov, Second-order sign-preserving conservative interpolation (remapping) on general grids, J. Comput. Phys. 184 (1) (2003) 266–298. doi:10.1016/S0021-9991(02)00033-5.

[9] F. Alauzet, M. Mehrenberger, P^1-conservative solution interpolation on unstructured triangular meshes, Int. J. Numer. Methods Eng. 84 (13) (2010) 1552–1588. doi:10.1002/nme.2951.

[10] F. Alauzet, A parallel matrix-free conservative solution interpolation on unstructured tetrahedral meshes, Comput. Methods Appl. Mech. Eng. 299 (2016) 116–142. doi:10.1016/j.cma.2015.10.012.

[11] T. Levý, G. May, Conservative solution transfer between anisotropic meshes for time-accurate hybridized discontinuous Galerkin methods, Int. J. Numer. Methods Fluids 96 (6) (2024) 1011–1030. doi:10.1002/fld.5278.

[12] J. R. Maddison, C. J. Cotter, P. E. Farrell, Geostrophic balance preserving interpolation in mesh adaptive linearised shallow-water ocean modelling, Ocean Model. 37 (1–2) (2011) 35–48. doi:10.1016/j.ocemod.2010.12.007.

[13] J. R. Maddison, C. J. Cotter, P. E. Farrell, Corrigendum to "geostrophic balance preserving interpolation in mesh adaptive linearised shallow-water ocean modelling" [Ocean Model. 37 (2011) 35–48], Ocean Model. 68 (2013) 106. doi:10.1016/j.ocemod.2013.04.007.

[14] T. J. Barth, P. O. Frederickson, Higher order solution of the Euler equations on unstructured grids using quadratic reconstruction, in: 28th Aerospace Sciences Meeting, AIAA Paper 90-0013, Reno, Nevada, 1990.

[15] T. J. Barth, Recent developments in high order k-exact reconstruction on unstructured meshes, in: 31st Aerospace Sciences Meeting, AIAA Paper 93-0668, Reno, Nevada, 1993.

[16] D. J. Mavriplis, Revisiting the least-squares procedure for gradient reconstruction on unstructured meshes, in: 16th AIAA Computational Fluid Dynamics Conference, Orlando, Florida, 2003, AIAA Paper 2003-3986. doi:10.2514/6.2003-3986.

[17] B. Diskin, J. L. Thomas, Accuracy of gradient reconstruction on grids with high aspect ratio, Tech. Rep. NIA Report 2008-12, National Institute of Aerospace, Hampton, Virginia (2008).
[18] D. A. Dunavant, High degree efficient symmetrical Gaussian quadrature rules for the triangle, Int. J. Numer. Methods Eng. 21 (6) (1985) 1129–1148.
[19] P. Keast, Moderate-degree tetrahedral quadrature formulas, Comput. Methods Appl. Mech. Eng. 55 (1986) 339–348.
[20] T. J. Barth, D. C. Jespersen, The design and application of upwind schemes on unstructured meshes, in: 27th Aerospace Sciences Meeting, AIAA Paper 89-0366, Reno, Nevada, 1989. doi:10.2514/6.1989-366.
[21] M. Shashkov, B. Wendroff, The repair paradigm and application to conservation laws, J. Comput. Phys. 198 (1) (2004) 265–277.
[22] M. Kucharik, M. Shashkov, B. Wendroff, An efficient linearity-and-bound-preserving remapping method, J. Comput. Phys. 188 (2) (2003) 462–471.
[23] S. T. Zalesak, Fully multidimensional flux-corrected transport algorithms for fluids, J. Comput. Phys. 31 (3) (1979) 335–362.
[24] B. Koobus, C. Farhat, Second-order time-accurate and geometrically conservative implicit schemes for flow computations on unstructured dynamic meshes, Comput. Methods Appl. Mech. Eng. 170 (1–2) (1999) 103–129. doi:10.1016/S0045-7825(98)00207-2.
[25] P. Geuzaine, C. Grandmont, C. Farhat, Design and analysis of ALE schemes with provable second-order time-accuracy for inviscid and viscous flow simulations, J. Comput. Phys. 191 (1) (2003) 206–227. doi:10.1016/S0021-9991(03)00311-5.
[26] M. B. Giles, N. A. Pierce, An introduction to the adjoint approach to design, Flow Turbul. Combust. 65 (2000) 393–415. doi:10.1023/A:1011430410075.
[27] R. J. LeVeque, Finite Volume Methods for Hyperbolic Problems, Cambridge Texts in Applied Mathematics, Cambridge University Press, Cambridge, 2002. doi:10.1017/CBO9780511791253.
[28] P. R. Spalart, S. R. Allmaras, A one-equation turbulence model for aerodynamic flows, Rech. Aérosp. (1) (1994) 5–21.
[29] T. D. Economon, F. Palacios, S. R. Copeland, T. W. Lukaczyk, J. J. Alonso, SU2: An open-source suite for multiphysics simulation and design, AIAA J. 54 (3) (2016) 828–846. doi:10.2514/1.J053813.
[30] F. R. Menter, Two-equation eddy-viscosity turbulence models for engineering applications, AIAA J. 32 (8) (1994) 1598–1605. doi:10.2514/3.12149.